# Learning Transferable Basin and River-Network Representations for Hydrologic Prediction Across Space and Time

How we represent a basin decides whether a model generalizes


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| | |
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- Conclusion

# Academic Background

**University of Engineering and Technology, Lahore** (2017-2021)

- B.S. Computer Science

**Michigan State University** (2023-Present)

- Dual Ph. D.
  - Computer Science (advised by Dr. Pang-Ning Tan)
  - Biosystems and Agriculture Engineering (advised by Dr. A. Pouyan Nejadhashemi)

# Research Overview

# **Motivation:** Streamflow, what it is and why it matters

Streamflow is the **volume of water moving through a river** at a particular location over time.

- **Public safety**
  Flood warning, drought response, emergency planning
- **Agriculture & water supply**
  Irrigation demand, reservoir operation, water allocation
- **Ecosystems & infrastructure**
  Environmental flows, design standards, climate adaptation

NOAA reports losses of approximately $367.5 billion from drought and $203.0 billion from inland flooding over 1980–2024.

*Reliable streamflow information supports decisions under both **water excess** and **scarcity**.*

# **Motivation:** The gauge gap

**2.7 Million+**

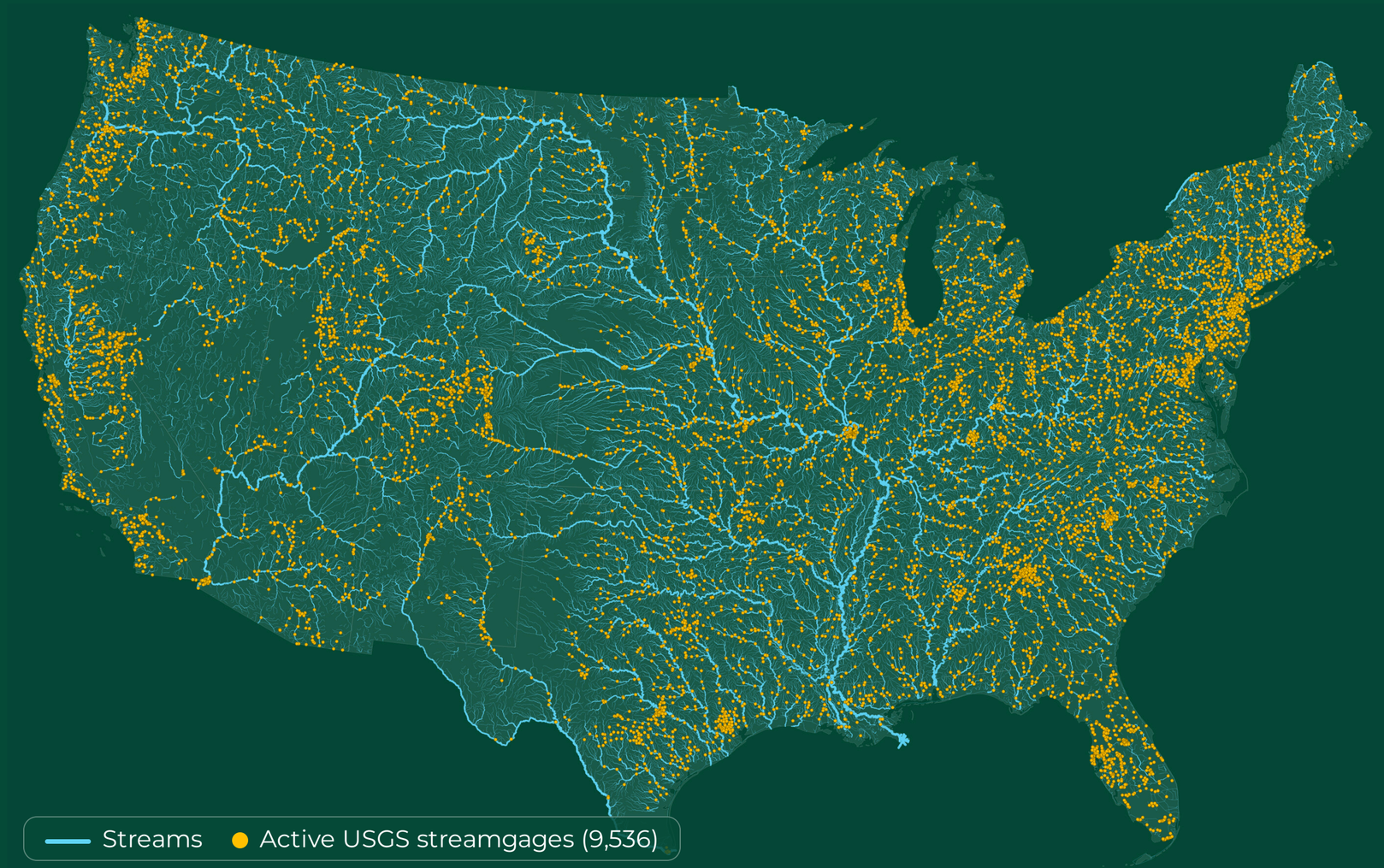
River reaches across the CONUS

**~12k**

Federal priority streamgage locations

**~5k**

Inactive gauges due to lack of funds




**Gauges cannot observe every reach. Models must predict where we do not measure.**

Rivers: USGS National Atlas 1:1M hydrography (lower 48). Gauges: USGS NWIS active stream sites reporting daily discharge, retrieved Nov 2026. Albers equal-area projection.

# **Monitoring is essential**!

but gauge measurements **cannot directly observe** every part of the national river network.

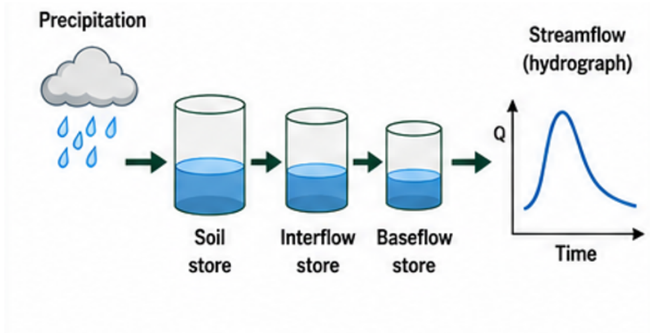
That's why **streamflow modeling** is needed

# How is streamflow modeled?

*Three broad model families represent the rainfall–runoff relationship in different ways.*

## 1. Conceptual models

*Simplified representation of watershed stores and fluxes.*




**Pros**

- Hydrologically interpretable
- Simple structure, efficient to run
- Works with relatively limited data

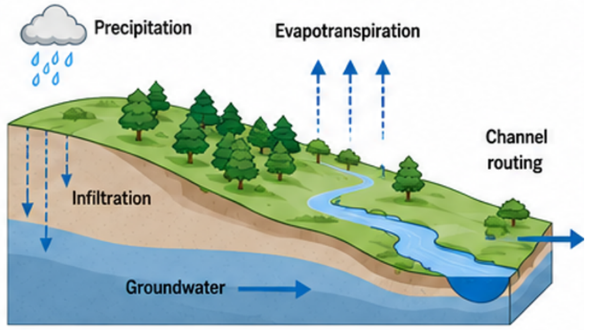
**Cons**

- Processes are simplified
- Requires calibration for each basin
- Limited ability to represent spatial variability

**Examples:** HBV, SAC-SMA, Xinanjiang, GR4J

## 2. Process-based models

*Explicit representation of physical processes across the watershed.*




**Pros**

- Physically based and interpretable
- Captures spatial and temporal processes in detail
- Can simulate many variables beyond streamflow

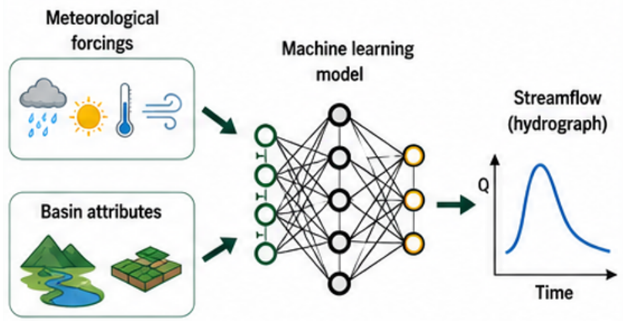
**Cons**

- Data intensive (inputs & parameters)
- Computationally expensive
- Difficult to calibrate and transfer across many basins

**Examples:** SWAT, VIC, MIKE SHE, HYPE, TOPKAPI

## 3. Data-driven models

*Learn the forcing–streamflow relationship from observed data.*




**Pros**

- Strong predictive performance
- Learns complex, nonlinear relationships
- Shares information across many basins (regional learning)

**Cons**

- Less interpretable
- Performance depends on quality of data and representation
- May extrapolate poorly outside training regimes

**Examples:** LSTM, EA-LSTM, GRU, Transformers, GNNs

**Takeaway:** Model family describes how the rainfall–runoff relationship is represented.
It is independent of the spatial setting (lumped vs. distributed) and the generalization task (temporal vs. spatial).

Modeling streamflow requires data.

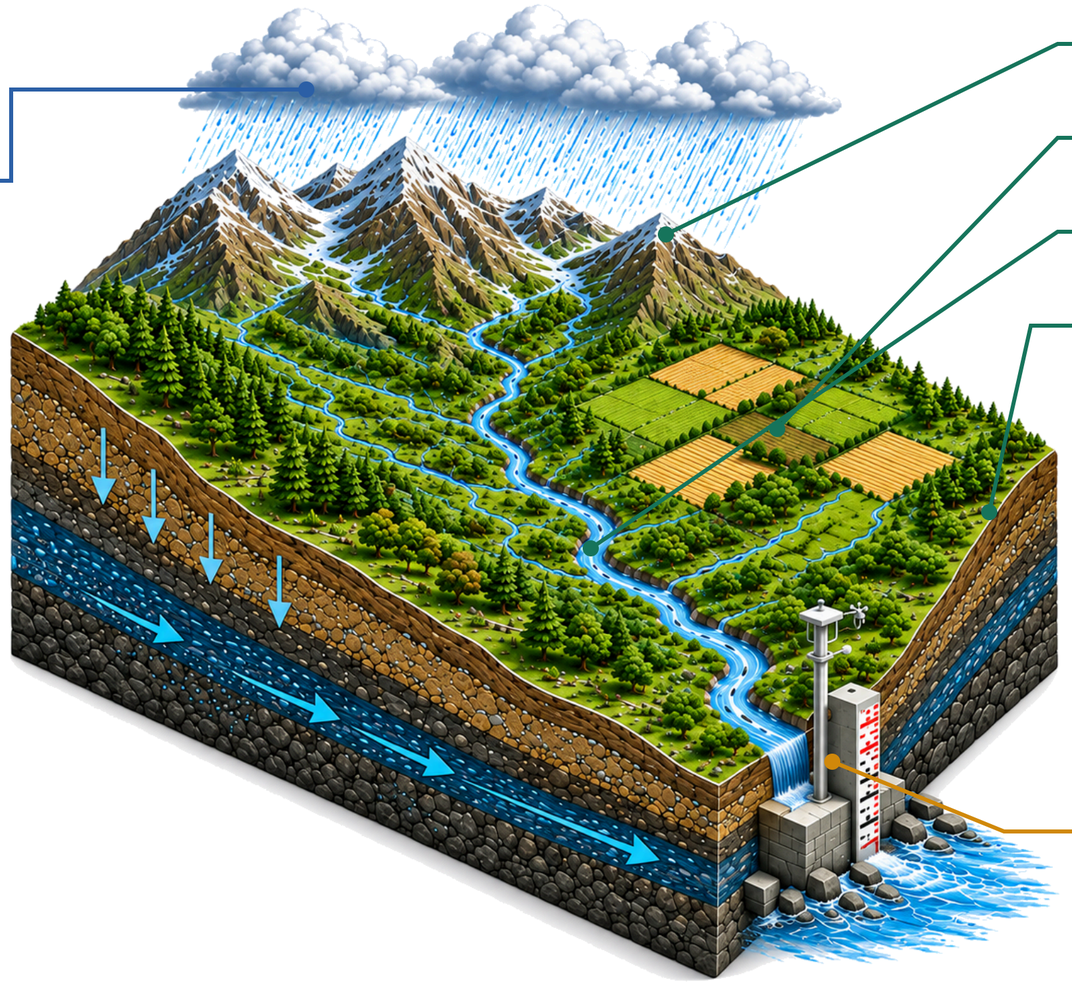
- Data types: **dynamic** and **static**

# Data Types

## DYNAMIC VARIABLES
(vary over time)

- **Precipitation**
  rainfall, snow
- **Temperature**
  min / max air temperature
- **Wind speed**
- **Humidity**
  vapor pressure
- **Air pressure**
- **Solar radiation**
  shortwave, day length

...

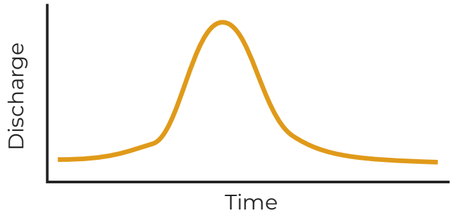


## STATIC VARIABLES
(do not vary over time)

- **Terrain**
  elevation, slope
- **Land cover**
  vegetation, crops
- **Drainage network**
  river topology
- **Soils**
  soil type, texture
- **Geology**
  rock type
- **Hydrologic properties**
  porosity, Ksat, AWC

...

## OBSERVED OUTPUT
Outlet discharge




Dynamic variables **drive** the hydrologic cycle
inputs that change over time

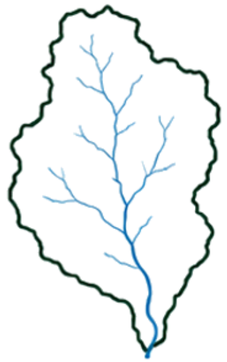
Static variables **define** catchment characteristics
inputs that remain constant

Modeling streamflow requires data.

- Data types: **dynamic** and **static**
- Data representation: **lumped** and **distributed**

# **Motivation:** Lumped vs. distributed basin representations

## Lumped Representation

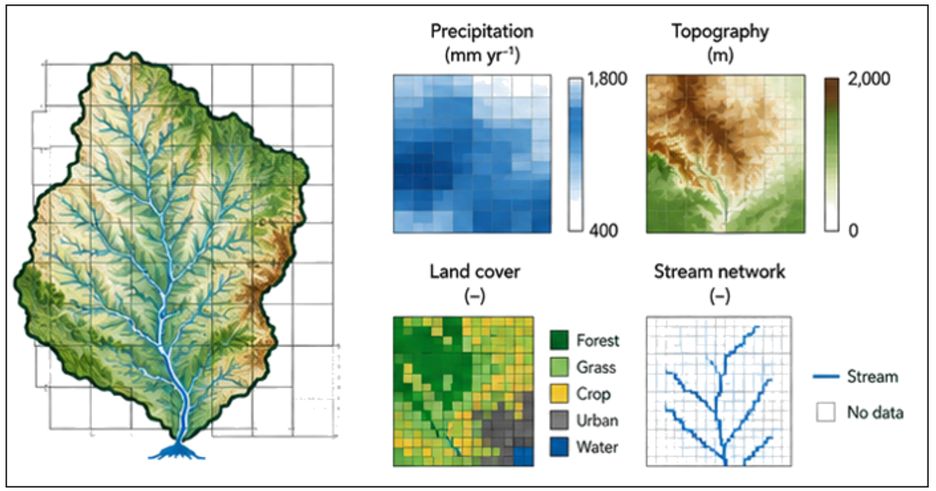


Aggregated attributes (feature vector)

| Feature | Example |
|---|---|
| Area ($km^2$) | 1,250 |
| Mean precipitation ($mm\ yr^{-1}$) | 1,120 |
| Mean slope (%) | 12.8 |
| Forest fraction (–) | 0.62 |
| Soil texture (sand %) | 42 |

**Trade-off**

## Distributed Representation



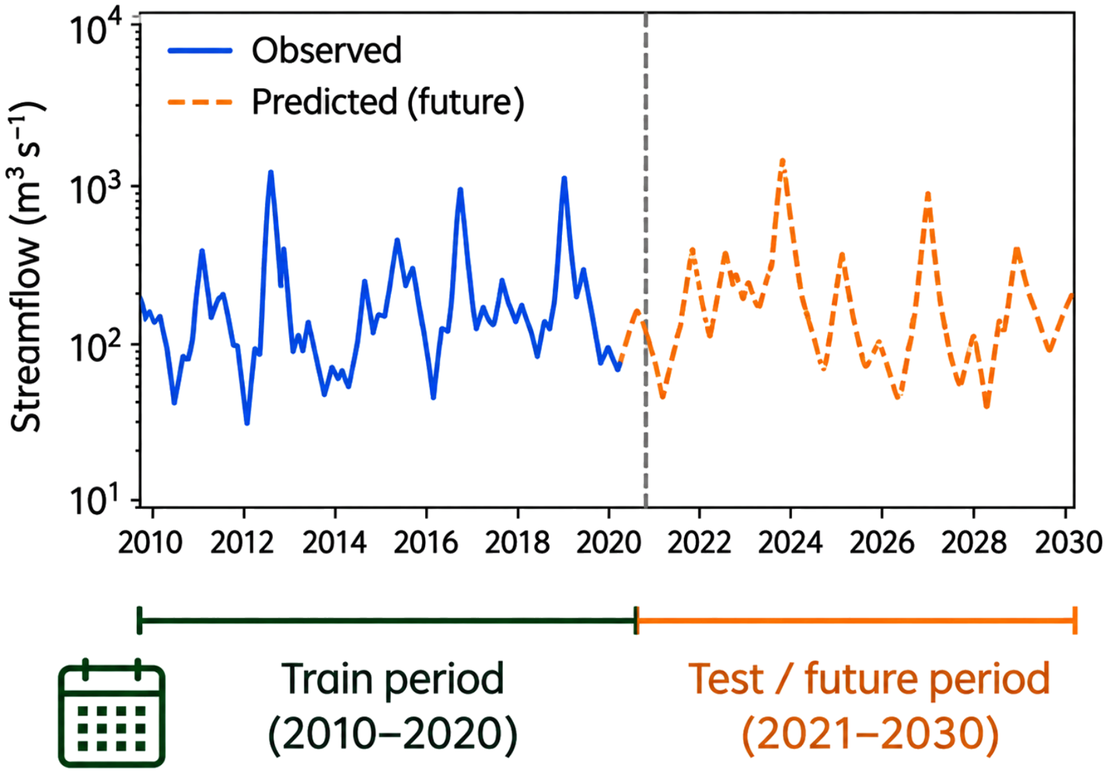

| Benefits | Limitations |
|---|---|
| • Compact<br>• Scalable<br>• Easy to use | • **Spatial heterogeneity lost**<br>• Local process information removed<br>• Drainage connectivity simplified |

| Benefits | Limitations |
|---|---|
| • **Spatially rich**<br>• Retains basin geometry<br>• Keeps drainage structure | • Hard to align across sources<br>• Expensive to process and store<br>• **Hard to scale and standardize** |

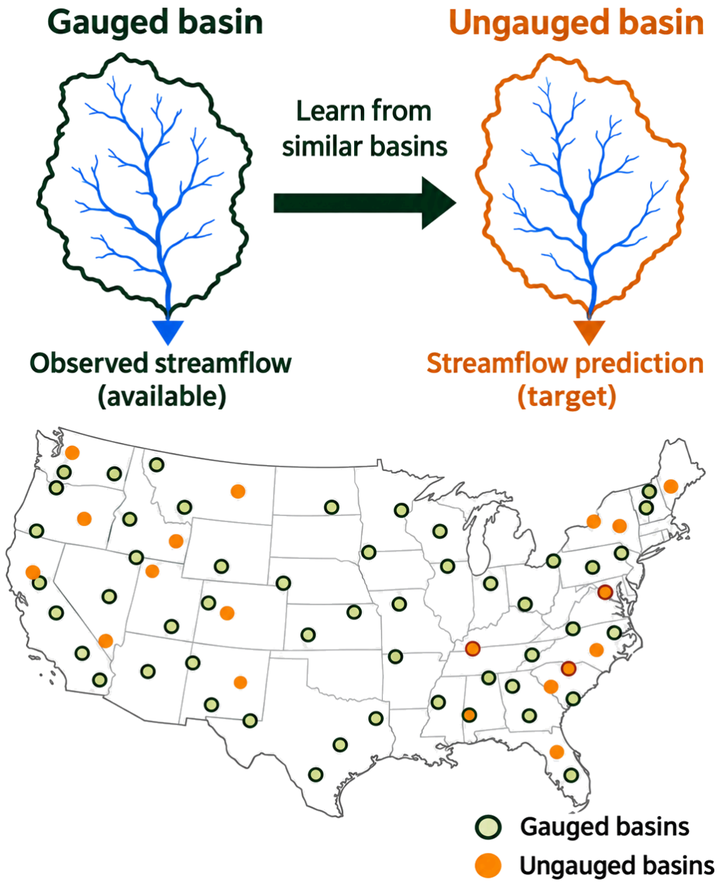
Representation must balance **compact** with **distributed spatial information**.

# **Motivation:** Two tests of generalization, time and space

## Temporal: same basins, future period




## Spatial: basins or gauges never seen




**Thesis: how we represent a basin decides whether a model generalizes.**

# **Literature Review:** What has been done, what limits it, what is missing

| | Hydrologic Modeling at Scale | Large-Sample Datasets | Transferability |
|---|---|---|---|
| **What has been done** | Mostly **lumped** basin representations<br>**LSTM, EA-LSTM**<br>*Kratzert et al. 2018, 2019; Addor et al. 2017* | Standardized basin-averaged forcing and catchment attributes<br>**CAMELS, Caravan**<br>*Addor et al. 2017; Linke et al. 2019; Kratzert et al. 2023* | Earth-observation models learn compact representations from imagery and geospatial data<br>**AlphaEarth, SatCLIP**<br>*Brown et al. 2025; Li et al. 2022; Bassi et al. 2024* |
| **Limitation** | **Internal spatial support removed.** Static and dynamic information combined by simple fusion. | **Distributed variables stay heterogeneous:** inconsistent resolution and projection across sources. | **Not hydrology-aware:** general representations ignore processes across basins, tasks and regimes. |
| **Gap** | 1 **Distributed models that preserve basin structure and fuse dynamic forcing with static variables better** | 2 **A standardized, basin-aligned dataset for large-scale distributed hydrologic learning** | 3 **Basin representations that keep hydrologic information and transfer across tasks, regions and climates** |

**Each gap sets up one of the studies that follow.**

# Selected Publications

| Title | Venue | Completion Date | Status |
|---|---|---|---|
| *Watershed-Agnostic Forecasting: An Explainable, Distributed, Data-Driven Hydrologic Model* | **Environmental Modelling & Software** | **Jun 2025** | **Published** |
| *From Aggregated to Distributed: A Transformer Framework for Generalizable Streamflow Modeling Across Space and Time* | **Engineering Applications of AI** | **Dec 2025** | **Revisions Submitted** |
| *BASINGRID: A distributed 40-year spatiotemporal hydrology dataset for U.S. basins* | **Nature Scientific Data** | **Apr 2026** | **Published** |
| *Basin2Vec: Learning Basin-Centric Embeddings for Scalable Hydrological Modeling* | **CVF WACV 2027** | **Jun 2026** | **Accepted** |

STUDY 1

# Fusion of Static and Dynamic Variables

HydroTransformer · the basin as a spatial field

# **Study 1:** Introduction

## Typical Lumped Modeling

**Dynamic variables $X_t$** (basin averages)

| P | Tmax | Tmin | SRad | VP | SWE |
|---|---|---|---|---|---|
| | | | | | |
| | | | | | |
| | | | | | |

⋮ one value per day

**Static attributes S**

| Elev | Slope | Clay | Sand | AWC | FC |
|---|---|---|---|---|---|
| | | | | | |

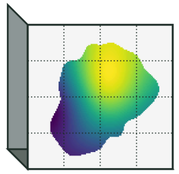
**[ $X_t$ ; S ]**
stacked as
extra channels

**LSTM** → *Q(t)*

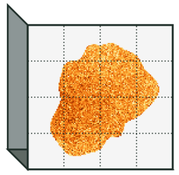
*Spatial organization is lost before the model sees the data.*

## Distributed Basin-Aware Modeling

**Dynamic variables $X_t$** (gridded)



**Static variables S**



**Spatial encoder**

✗ **No channel stacking**

**Conditional fusion**
static context
gates dynamic
features

→ *Q(t)*

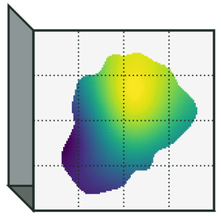
*Where things happen inside the basin is kept.*

**Study 1: condition distributed dynamic features on distributed static basin structure.**

# **Study 1:** Overview

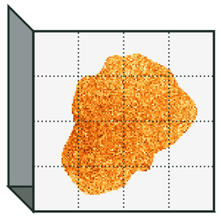
**QUESTION** Which variables are used to predict streamflow across time and in unseen basins?

## Dynamic Variables



1. Precipitation
2. Min and max temperature
3. Shortwave radiation
4. Vapor pressure
5. Daylength
6. Snow water equivalent

## Static Variables



1. Elevation
2. Clay, sand, and silt
3. Available water capacity
4. Field capacity

## Evaluation Settings

**Temporal**

future periods in observed basins

**Spatial**

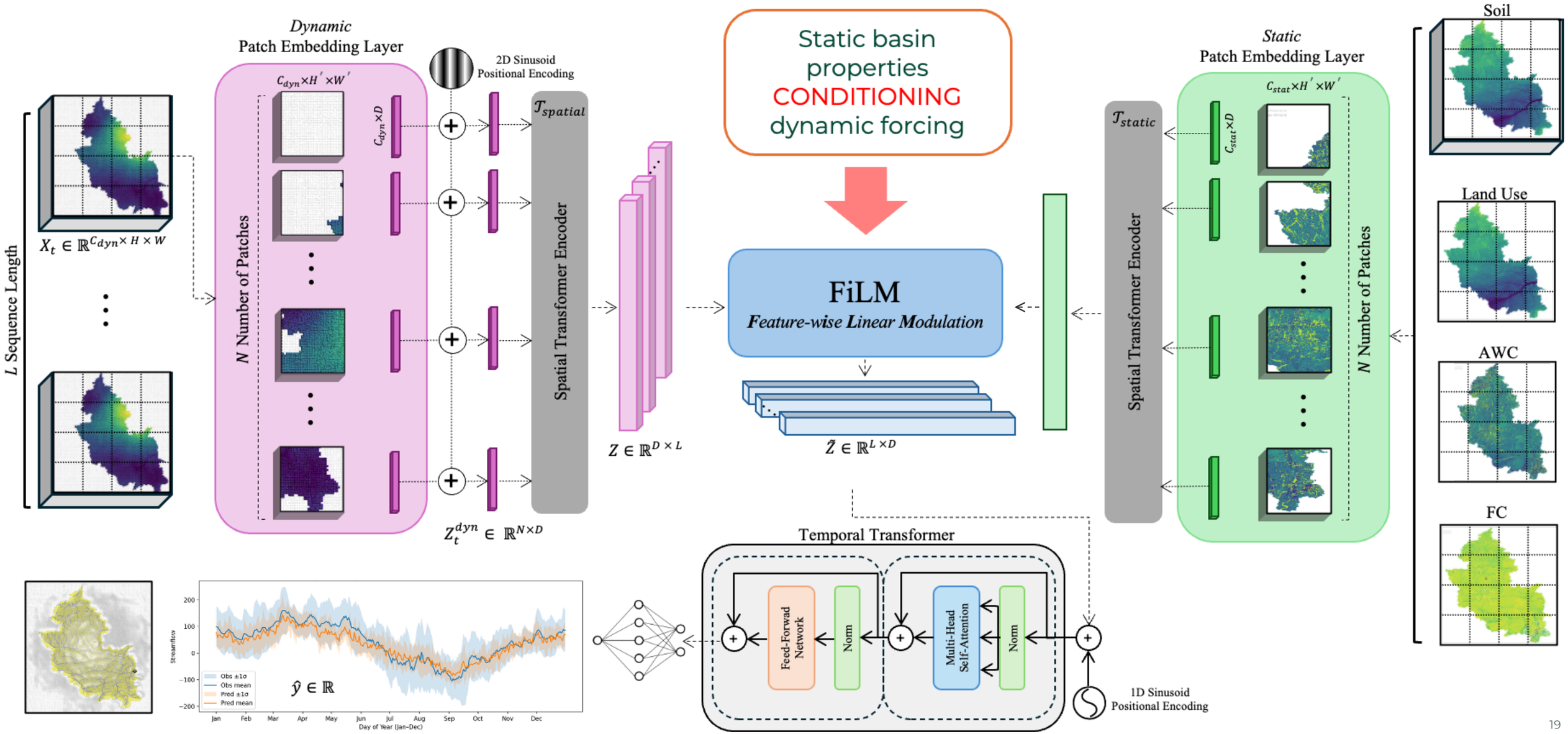
completely held-out basins

$$\hat{Q}_{b,t} = \mathrm{HydroTransformer}\left(X_{b,\, t-L+1:t},\ S_b\right)$$

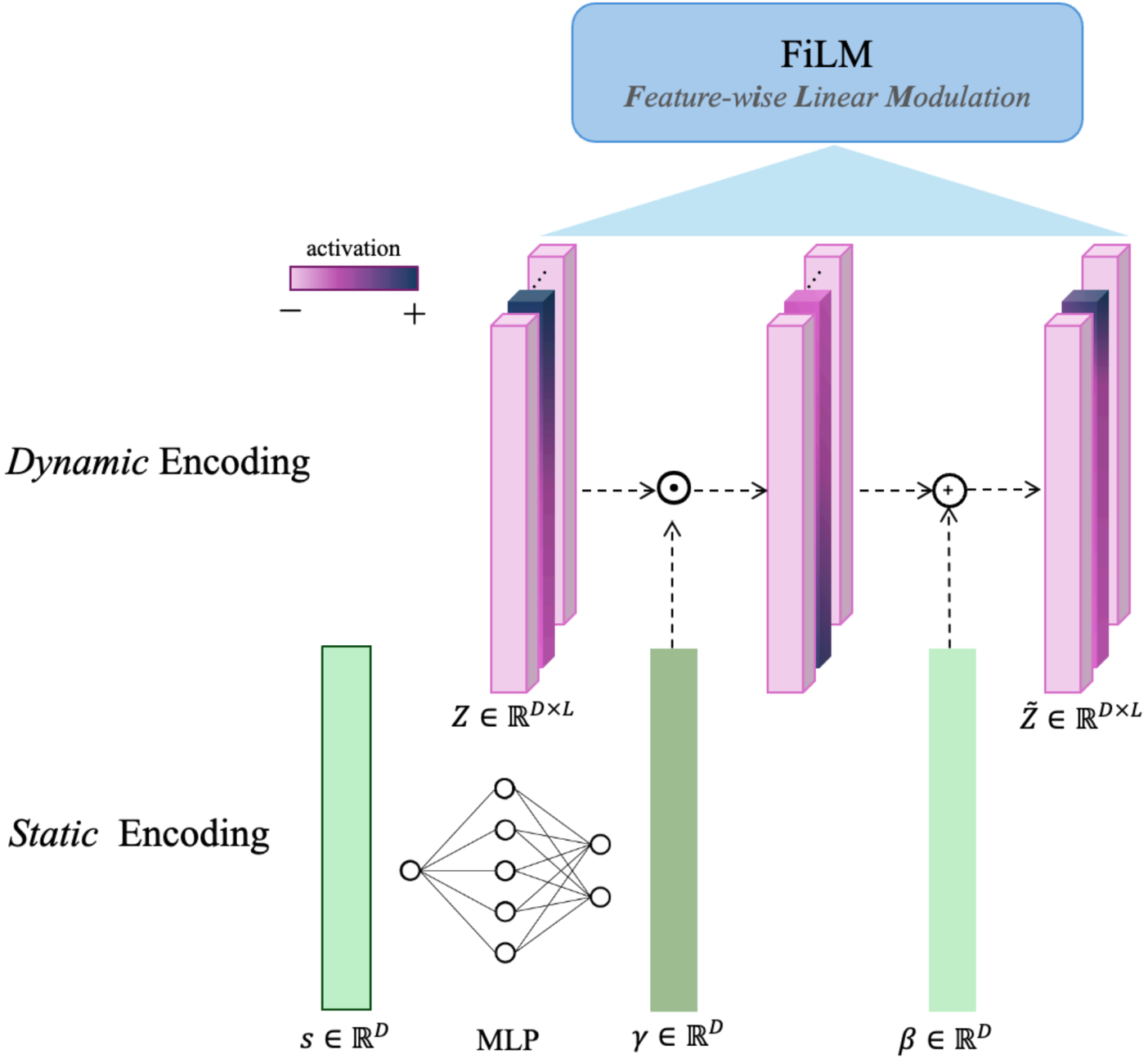
| | |
|---|---|
| $X_{b,\, t-L+1:t}$ | gridded dynamic variables over the last $L$ days |
| $S_b$ | gridded static variables of basin $b$ |
| $\hat{Q}_{b,t}$ | predicted discharge for basin $b$ on day $t$ |

# **Study 1:** HydroTransformer

Encode dynamic and static basin information separately, then integrate them through conditioning.



# **Study 1:** FiLM Conditioning




How static variables are CONDITIONING dynamic variables?

$$\tilde{z}_{b,t} = z_{b,t} \odot (1 + \gamma_b) + \beta_b$$

$\gamma_b$: Basin-specific **feature scaling**

$\beta_b$: Basin-specific **feature shifting**

# **Study 1:** Experimental setup

Does distributed basin structure improve generalization across time and unseen basins?

## Evaluation Settings

**Temporal generalization:** future periods in observed basins

**Spatial generalization:** completely held-out basins

### Benchmarks

LSTM · EA-LSTM · Informer · CNN-LSTM · HydroTransformer

## Generalization Tests

**Sequence Lengths:** {5, 21, 365}

**Temporal generalization**

**Train:** 2000–2013

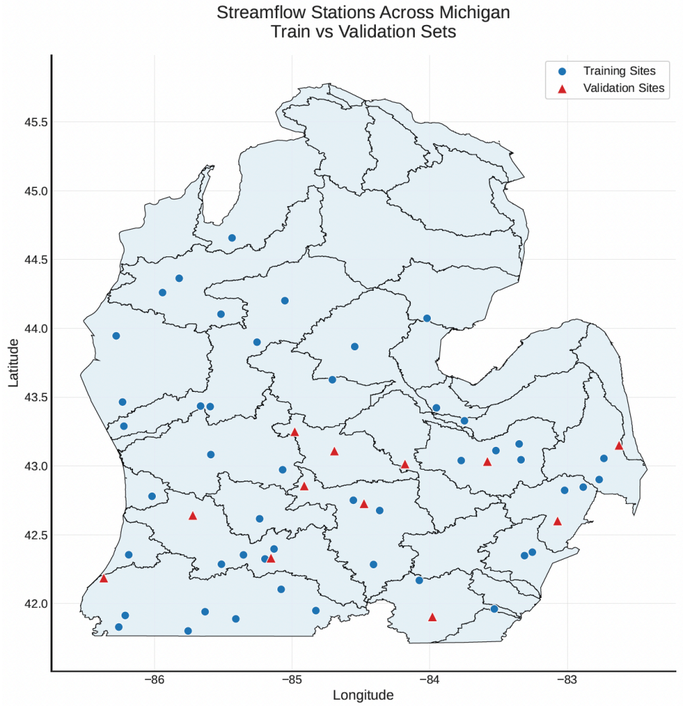
**Test:** 2014–2020

*Same basins, future period*

**Spatial generalization**

12 completely held-out basins

No discharge observations from test basins used during training




## Evaluation Metrics

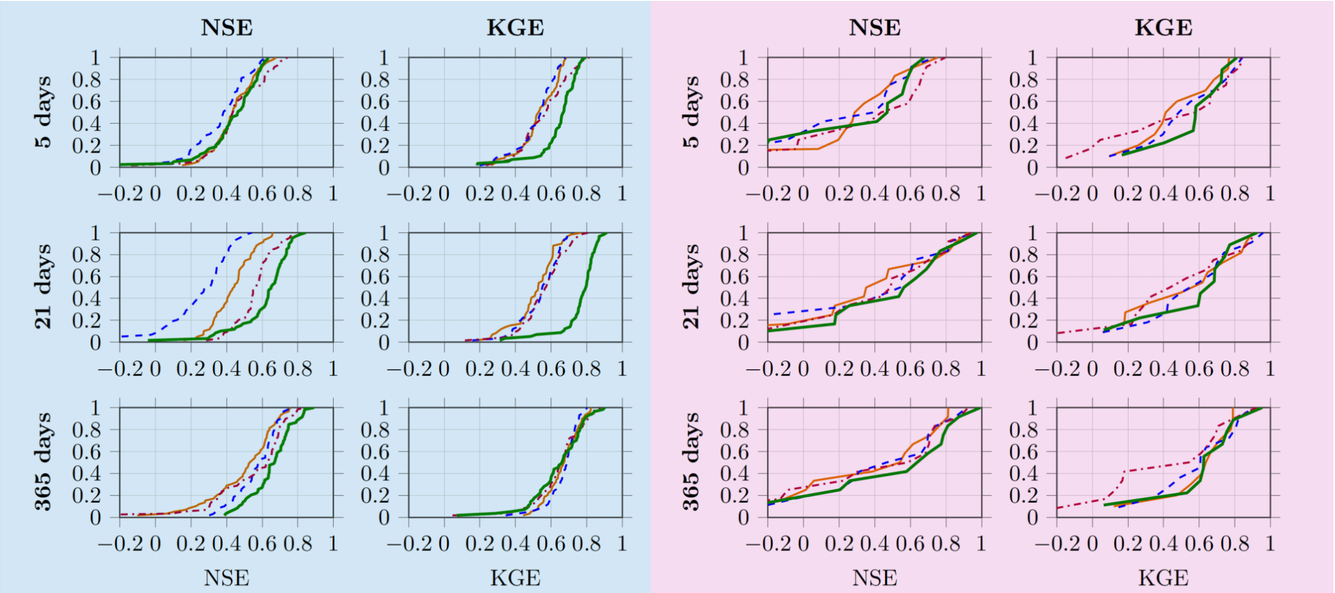
**Nash–Sutcliffe Efficiency (NSE)**
**Range:** (−∞, 1]

**Kling–Gupta Efficiency (KGE)**
**Range:** (−∞, 1]

# **Study 1:** Generalization Performance

With these two evaluation settings fixed, does distributed basin structure actually improve prediction?

## CDF Plots




(a) Temporal generalization performance distributions.




(b) Ungauged-basin generalization performance distributions.

LSTM  EA–LSTM  Informer  HydroTransformer

## NSE Evaluation

### Temporal Generalization

| | No static | | | With static | | |
|---|---|---|---|---|---|---|
| **Model** | 5 | 21 | 365 | 5 | 21 | 365 |
| LSTM | 0.180 | 0.063 | 0.119 | 0.439 | 0.444 | 0.535 |
| EA-LSTM | – | – | – | 0.383 | 0.309 | 0.579 |
| CNN–LSTM | 0.363 | 0.529 | **0.639** | – | – | – |
| Informer | 0.242 | 0.300 | 0.370 | 0.436 | 0.561 | 0.628 |
| HydroTransformer | **0.458** | **0.658** | 0.445 | **0.463** | **0.659** | **0.670** |

### Spatial Generalization

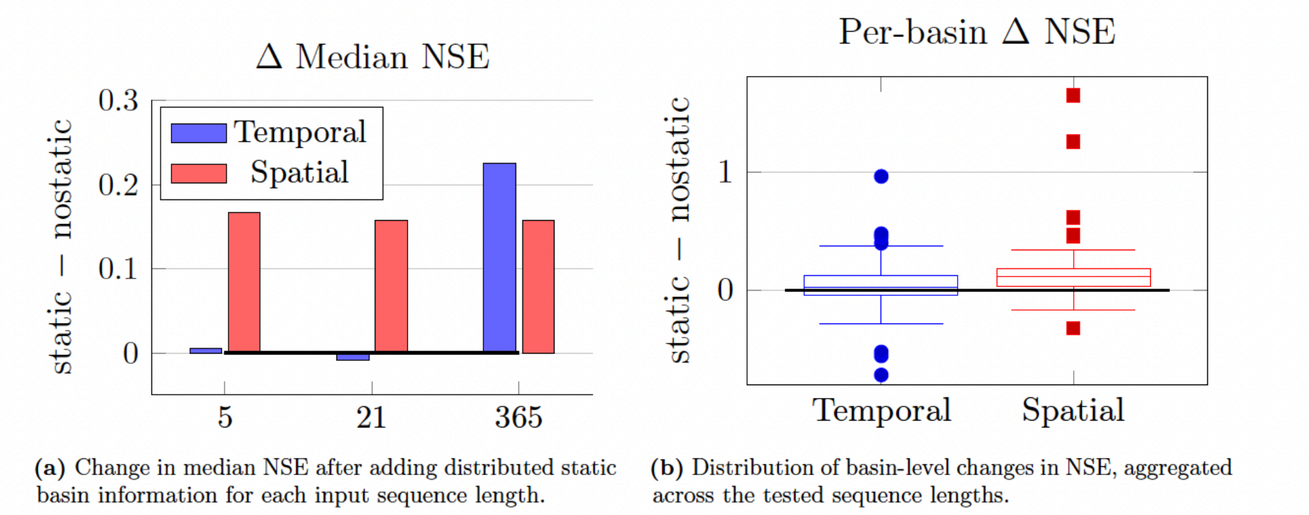
| | No static | | | With static | | |
|---|---|---|---|---|---|---|
| **Model** | 5 | 21 | 365 | 5 | 21 | 365 |
| LSTM | 0.273 | 0.455 | 0.590 | 0.314 | 0.408 | 0.555 |
| EA-LSTM | – | – | – | 0.420 | 0.547 | 0.563 |
| CNN–LSTM | 0.295 | 0.444 | 0.595 | – | – | – |
| Informer | **0.574** | **0.554** | **0.609** | **0.512** | 0.493 | 0.634 |
| HydroTransformer | 0.304 | 0.439 | 0.510 | 0.470 | **0.596** | **0.668** |

**HydroTransformer** improves generalization, most clearly in ungauged basins.

# **Study 1:** Impact of Distributed Static Attributes

How does static basin information improve HydroTransformer's performance?

## Performance Gain




**(a)** Change in median NSE after adding distributed static basin information for each input sequence length.

**(b)** Distribution of basin-level changes in NSE, aggregated across the tested sequence lengths.

$$\Delta\text{NSE} = \text{NSE}_{\textit{with static}} - \text{NSE}_{\textit{without static}}$$

**Temporal**

$$\Delta\text{NSE} = \begin{cases} +0.005, & 5d \\ +0.001, & 21d \\ +0.225, & 365d \end{cases}$$

**Spatial**

$$\Delta\text{NSE} = \begin{cases} +0.166, & 5d \\ +0.157, & 21d \\ +0.158, & 365d \end{cases}$$

**Distributed static variables** improve generalization.

# **Study 1:** Conclusion

**Study 1 shows that keeping the basin as a spatial field improves generalization.**

Established

- **Distributed fields** carry more information than basin-averaged (lumped) inputs.
- **Conditioning** dynamic fields on static basin structure outperforms simple concatenation.
- The gains are largest in **unseen basins**.

Next Challenge

- The workflow is region-specific: it was built for 61 basins in Michigan.
- Distributed rasters are computationally expensive to process, which limited us to six static variables.
- Scaling to thousands of basins requires standardized data.

**Next: BASINGRID, a standardized dataset**

STUDY 2

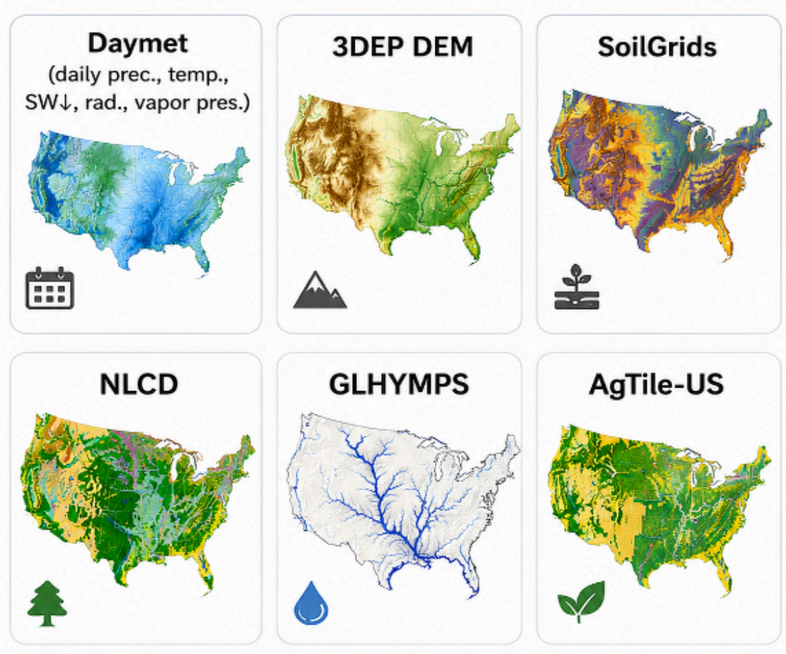
# BASINGRID

A distributed 40-year spatiotemporal hydrology dataset for U.S. basins

# **Study 2:** Motivation

Trade-off of scale and spatial structure

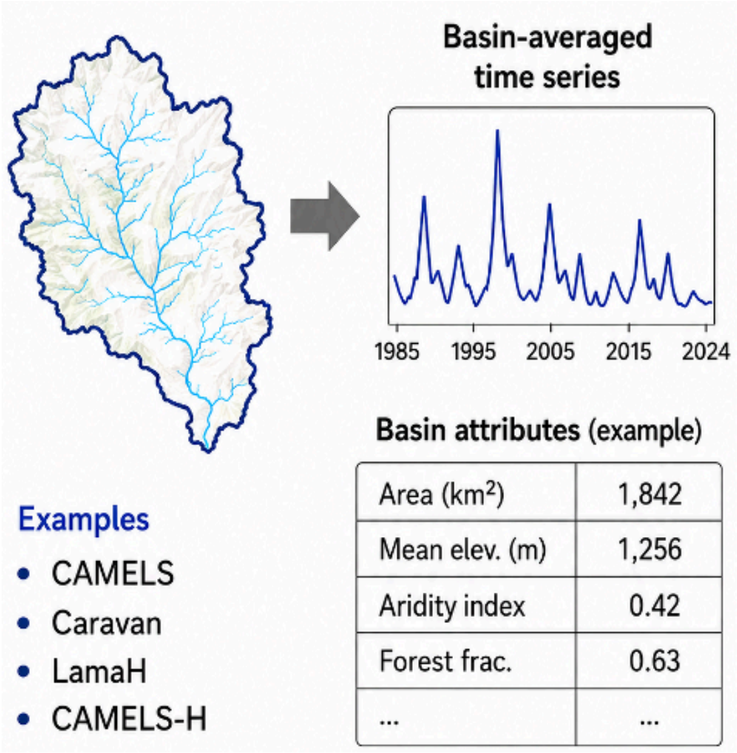
## Raw gridded data sources




Different projections, resolutions and coverage

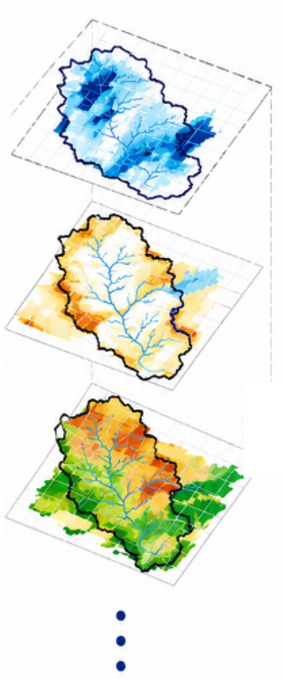
**Heterogeneity**

## Existing large-scale datasets




**Loss of spatial structure**

## What do we need?



**Preserve spatial information and make the data scalable**

# **Study 2:** Variables and data sources

**16 variables** at three native temporal resolutions

## DAILY

365 time steps / year
**6 variables**

### Meteorological forcings

Precipitation · minimum and maximum temperature · shortwave radiation · snow water equivalent · vapor pressure

**SOURCE**

Daymet: ~1 km gridded daily meteorology

## ANNUAL

1 time step / year
**2 variables**

### Land-surface evolution

Potential evapotranspiration (PET) · land-cover class

PET is computed daily with Hargreaves and stored as an annual sum.

**SOURCES**

PET: derived from CW3E
Land cover: NLCD

## STATIC

no time dimension
**8 variables**

### Basin attributes

**Topography:** elevation · **Soil:** clay, silt, sand, bulk density
**Hydrogeology:** permeability, hydraulic conductivity ·
**Agriculture:** tile drainage

**SOURCES**

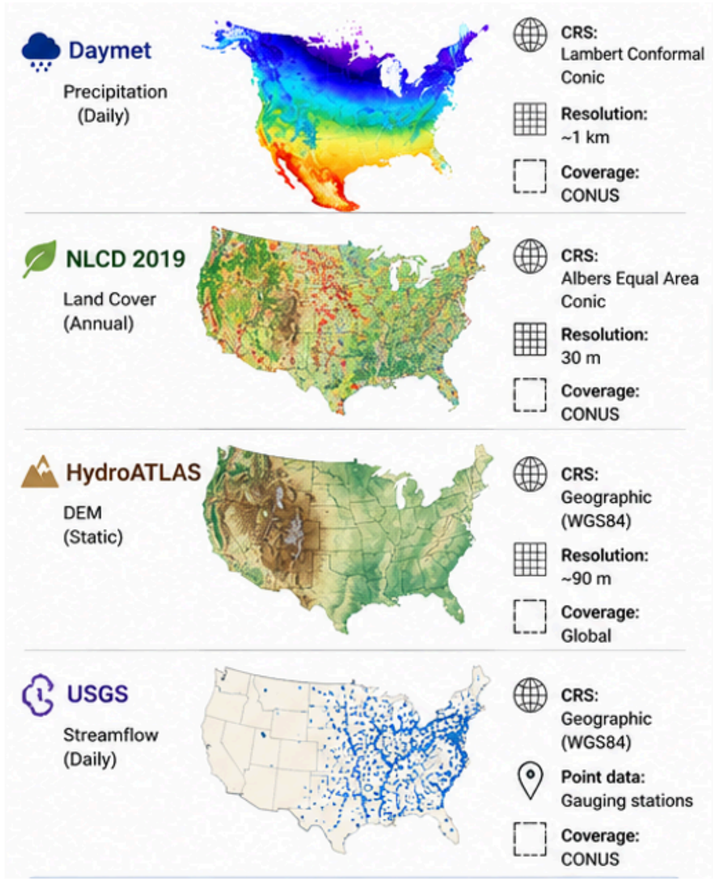
DEM: USGS 3DEP · Soil: SoilGrids
Subsurface: GLHYMPS · Tile drainage: AgTile-US

**Different coordinate reference systems + different spatial and temporal resolutions**

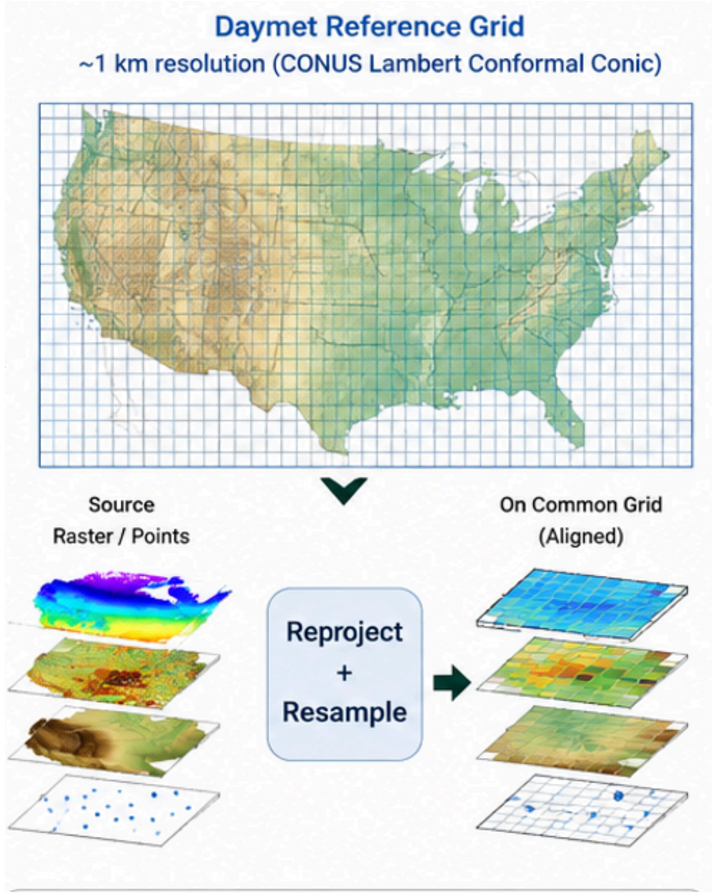
# **Study 2:** A common grid for every basin

**Different sources** (CRS, resolution, coverage) ⟶ **one reference grid** ⟶ **basin-aligned 64 × 64**

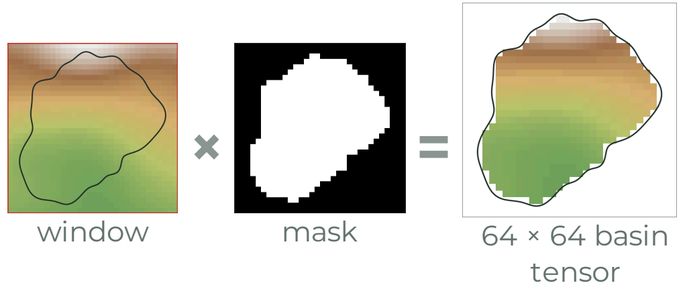
## Heterogeneous source grids




## Common grid




## Basin-aligned output




values + mask + physical-scale metadata

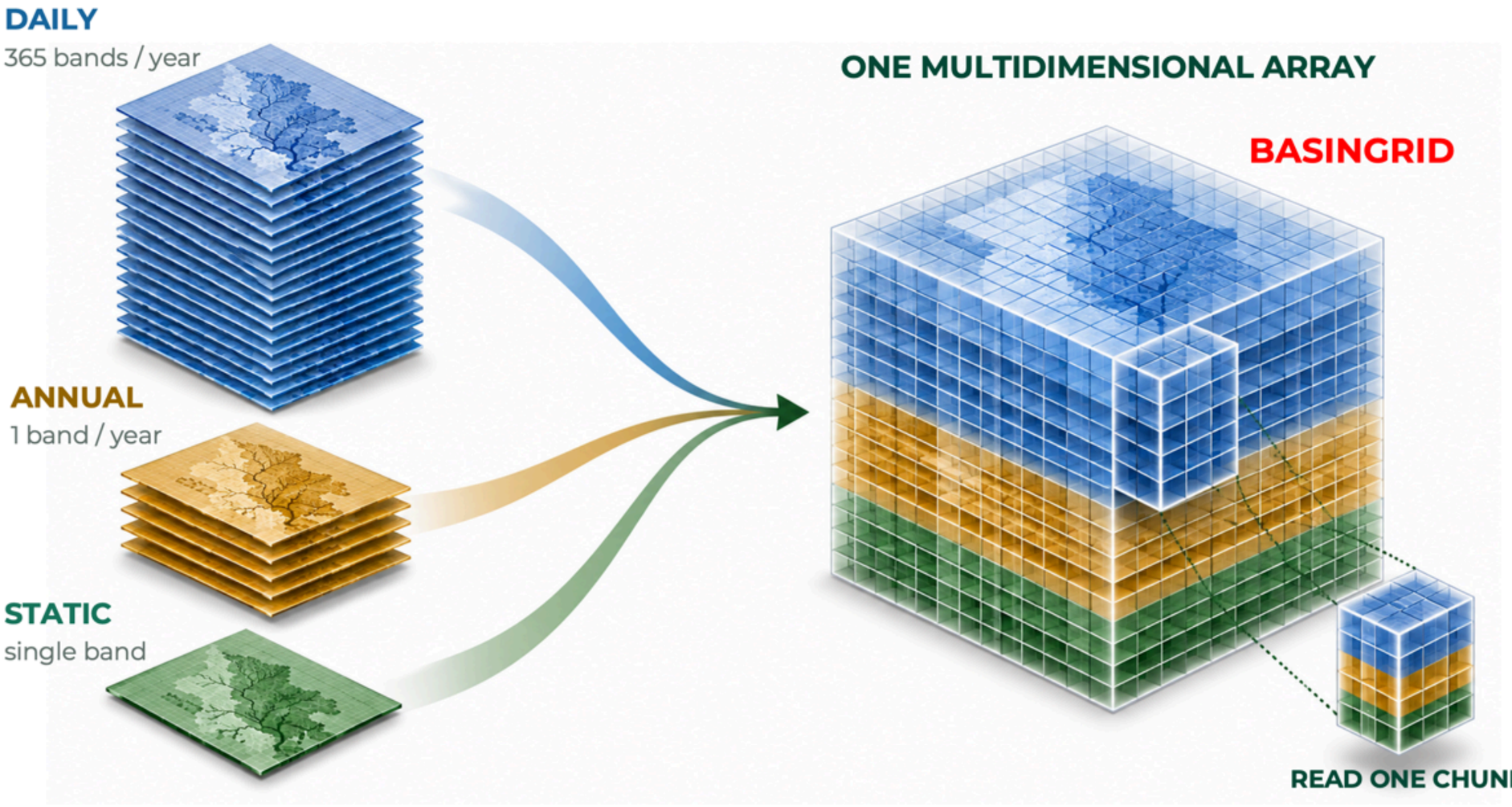
**Continuous** variables: bilinear interpolation

**Categorical** variables: nearest neighbor

**Standardized CRS · aligned grid · consistent tensor shape**

# **Study 2:** Organizing the Data Cube

**Native temporal resolutions are preserved within one model-ready array structure.**




## ZARR STORAGE

**Designed for selective, large-scale access**

- **CHUNKED**
  Basin and temporal blocks are accessed independently.
- **COMPRESSED**
  Blosc with Zstandard reduces storage demand.
- **SCALABLE**
  Parallel workers and resumable execution.

ARRAY INDEX $BASIN \times YEAR \times TIME \times VARIABLE \times 64 \times 64$

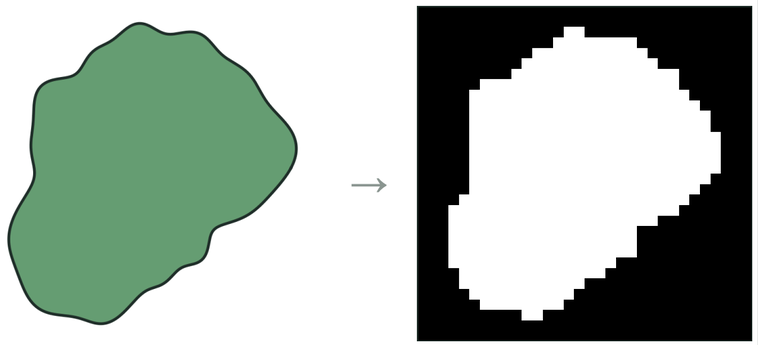
**PRESERVE TEMPORAL RESOLUTION · STANDARDIZE SPATIAL SHAPE · SCALE ACCESS ACROSS 9,067 BASINS**

# **Study 2:** Validation metrics

Three complementary checks: **basin area**, **continuous values**, and **categories**

## Area preservation

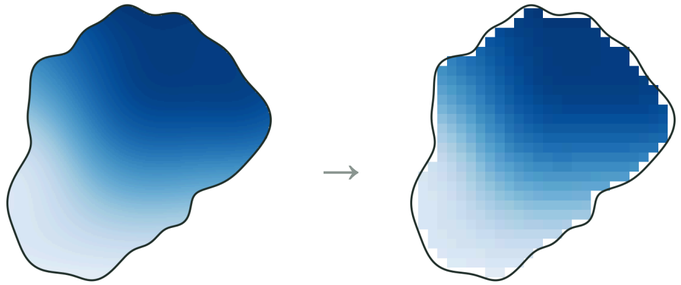
Does the normalized mask keep the basin area?



$R_A = A_{\text{mask}} / A_{\text{vector}}$
**Ideal:** $R_A = 1$, same physical area

## Continuous values

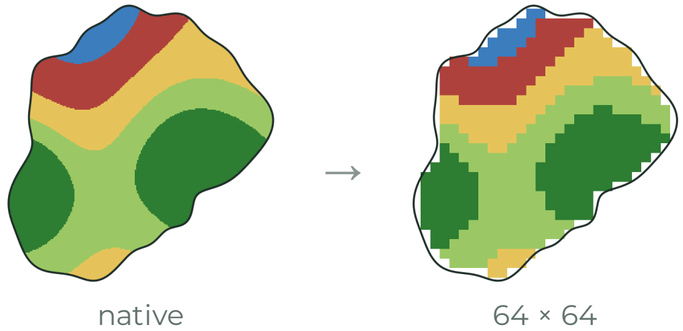
Does the patch keep continuous values?




**MdAPE**: typical % difference from the native raster
**Ideal:** MdAPE → 0%

## Categorical fidelity

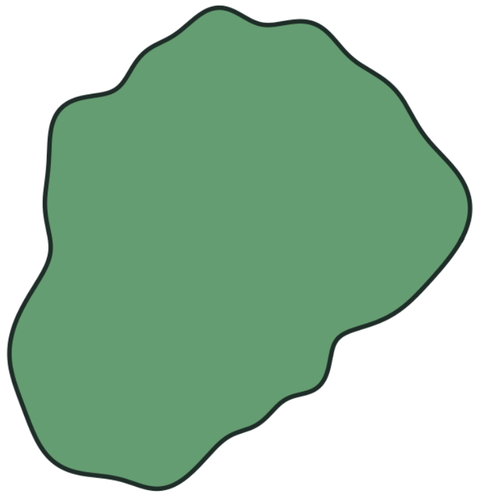
Are land-cover categories kept?




**Class-proportion distance** and **dominant-class agreement**
**Ideal:** 0% distance, 100% agreement

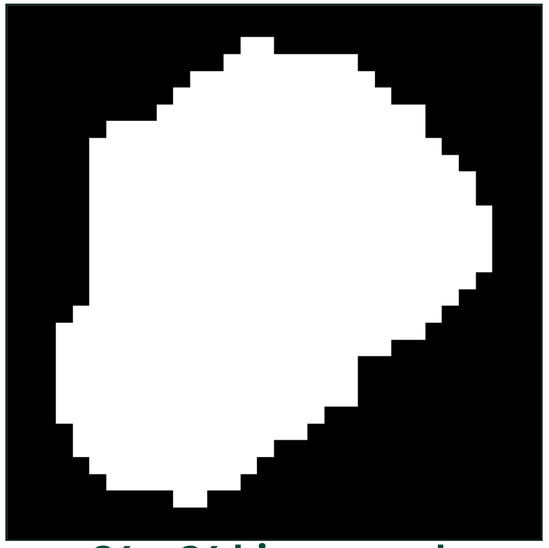
# **Study 2:** Validation, basin area is preserved

After restoring basin-specific pixel area, the normalized mask closely matches the original vector basin.



**Original vector basin**

→



**64 × 64 binary mask**

Area agreement · 9,067 basins

**1.094**

median mask / vector area ratio

**1.100**

mean mask / vector area ratio

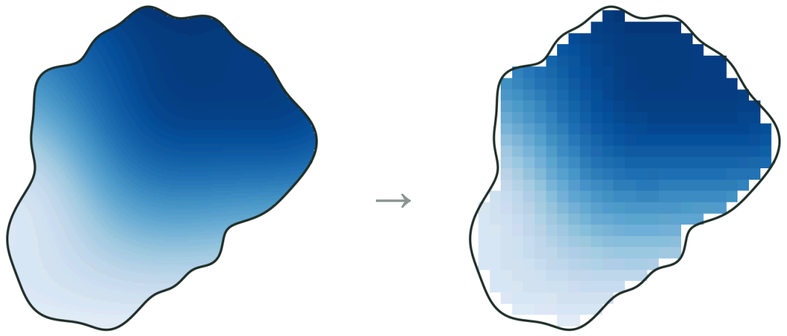
Normalization changes the **effective resolution**, not the basin's **physical area**.

The 64 × 64 normalization keeps basin-size information.

# **Study 2:** Validation, resampled values

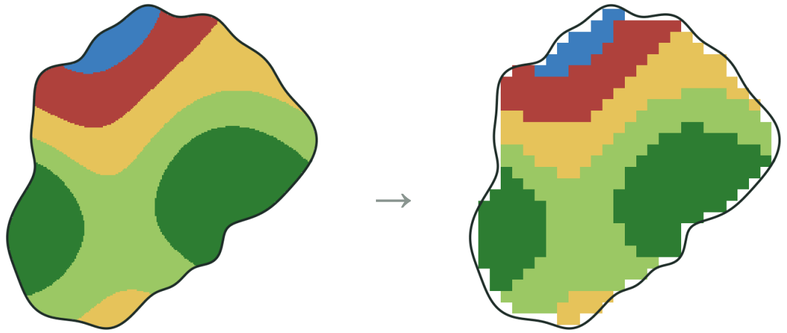
Native-grid and 64 × 64 patch values stay closely aligned across small, medium, and large basins.

**Continuous: precipitation**



native grid → 64 × 64 patch

**Categorical: land cover**



native grid → 64 × 64 patch

### Precipitation fidelity

**< 1%**

MdAPE in every basin-size class

### Land-cover fidelity

**100%**

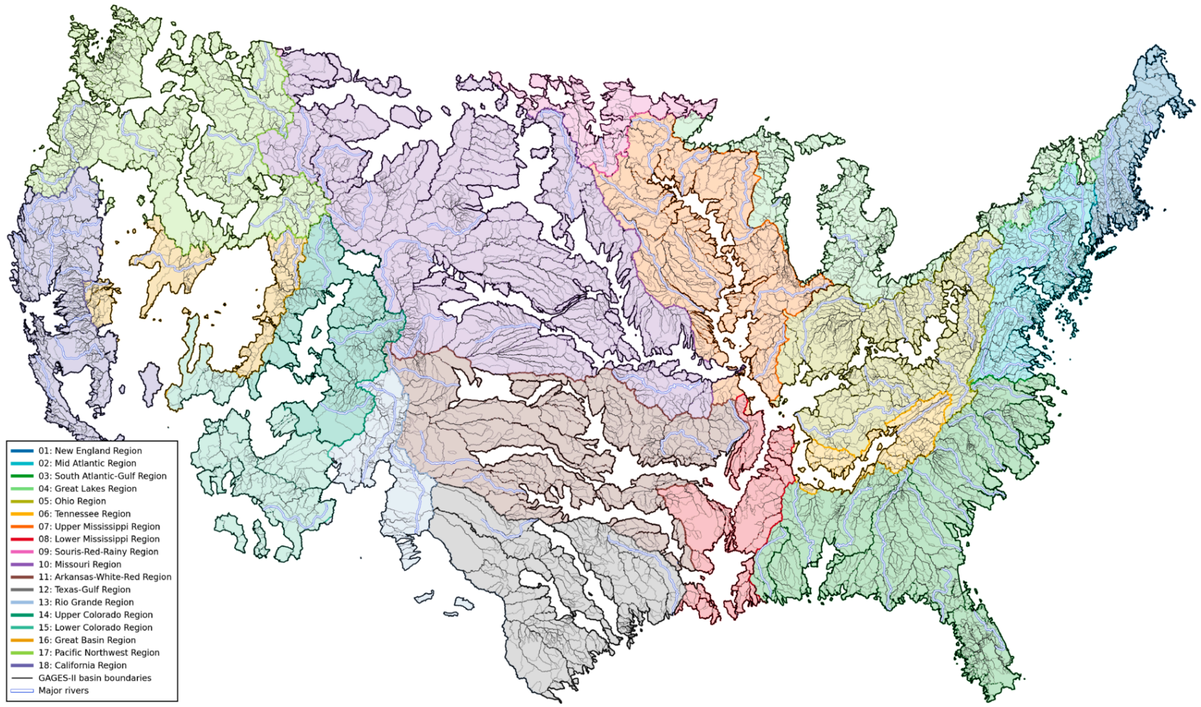
dominant-class agreement

class-proportion distance 1.23–2.89%

The 64 × 64 representation keeps basin-scale information across basin sizes.

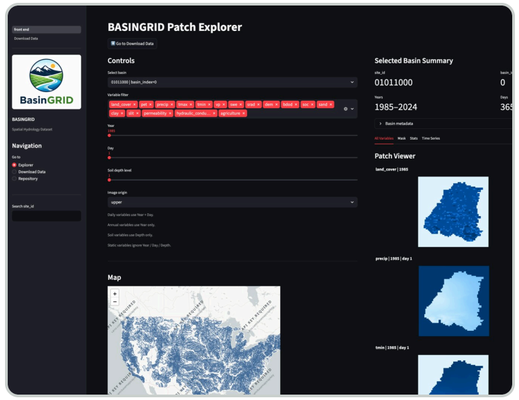
# **Study 2:** BASINGRID

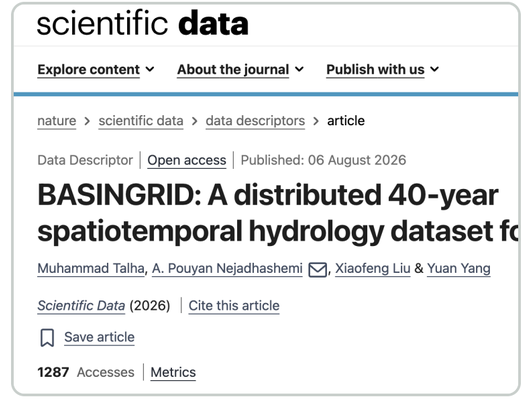
## Spatial coverage




## Dataset

| | |
|---|---|
| **Spatial scale** | **9,067** GAGES-II basins, CONUS |
| **Time period** | **1985–2024** |
| **Resolution** | Daily · 64 × 64 per basin |
| **Format** | ML-ready Zarr chunks |
| **Variables** | 16 · 6 daily, 2 annual, 8 static |




**basingrid.ciwre.msu.edu** · Scientific Data

BASINGRID turns distributed basin modeling from a study-specific workflow into reusable infrastructure.

# **Study 2:** Conclusion

**Study 2 shows that the spatial-field representation can be standardized and scaled to the national level.**

Established

- Heterogeneous rasters can be harmonized into **common basin-aligned tensors**.
- The 64 × 64 patches preserve basin **geometry, values, and categories**.
- The pipeline scales to **9,067 basins** over **1985–2024**, using chunked, parallel storage.

Next Challenge

- Each basin is now a stack of variables × time × 64 × 64 values.
- Every downstream model would have to learn to read this stack from scratch.
- The tensors are **too high-dimensional** to reuse directly.

**Next: Basin2Vec, a compact basin vector**

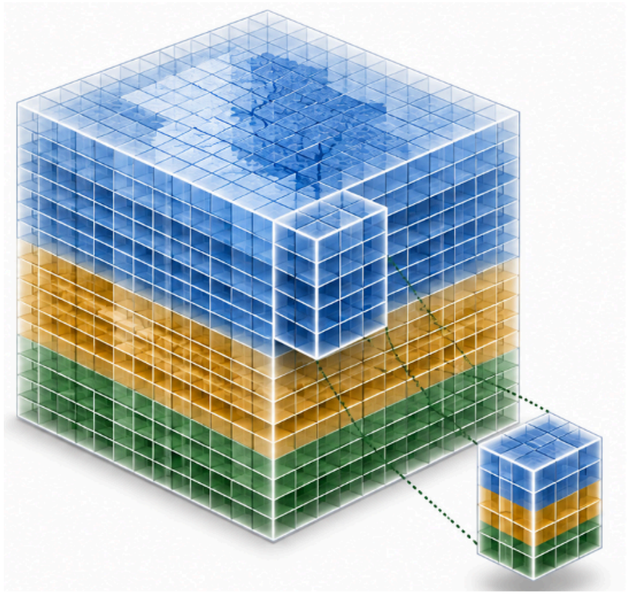
STUDY 3

# Basin2Vec

Learning basin-centric embeddings for scalable hydrologic modeling

# **Study 3:** Motivation

## What we have now



**BASINGRID:** dynamic static annual basin mask

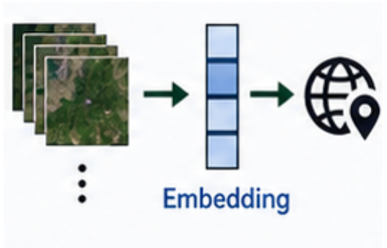
**Spatially rich, but high-dimensional for downstream models**

## Current compact alternatives

| $f_1$ | $f_2$ | $f_3$ | ⋯ | $f_p$ |
|---|---|---|---|---|
| 0.82 | -1.34 | 0.57 | ⋯ | 1.11 |

Feature vector

**A.** CAMELS attributes

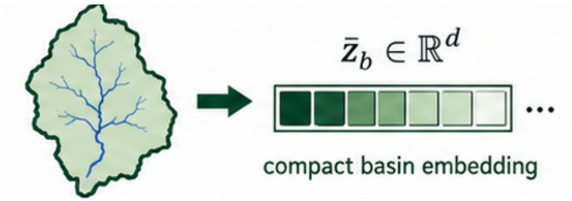



**B.** EO embeddings

**A: loses spatial structure**

**B: not hydrology-aware**

## What do we need?




One compact vector per basin

**Learned embeddings from hydrology data**

# **Study 3:** Why existing representations fall short

## Comparison

| Approach | Basin-indexed | Spatial & scale-aware | Hydrologic relation-aware |
|---|---|---|---|
| CAMELS attributes | ●●● | ●●● | ●●● |
| AlphaEarth | ●●● | ●●● | ●●● |
| TESSERA | ●●● | ●●● | ●●● |
| SatCLIP | ●●● | ●●● | ●●● |
| **Basin2Vec (ours)** | ●●● | ●●● | ●●● |

●●● High ●●● Moderate ●●● Low

Requirement 1

### Basin-indexed

One representation for one basin.

Requirement 2

### Spatial and scale-aware

Learns within-basin structure and keeps physical basin size.

Requirement 3

### Hydrologic relation-aware

Embedding geometry reflects hydrologic relations.

**Goal:** a compact, basin-indexed representation whose geometry is organized by hydrologic relations.

# **Study 3:** Methodology overview

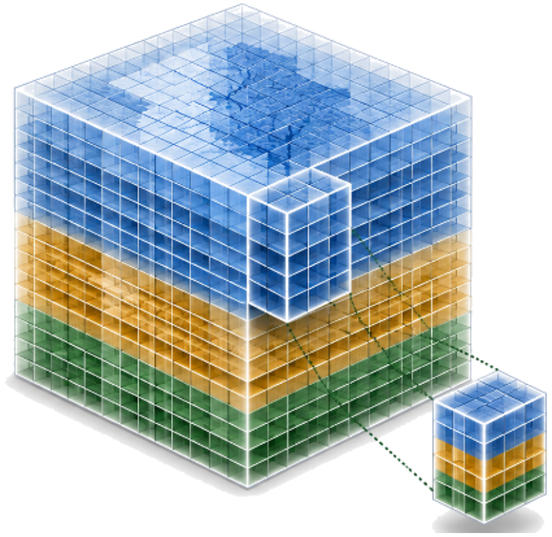
From a **basin-year tensor** to a **reusable basin embedding**

## Input entity

$(b, y)$

one basin-year

## Basin-year tensor



**Monthly dynamic** · **annual** · **static**

**Basin mask** + **metadata**, from BASINGRID

## Basin2Vec

**Scale-aware encoder**

mask-aware spatial encoding of the tensor

**Relation-aware self-supervision**

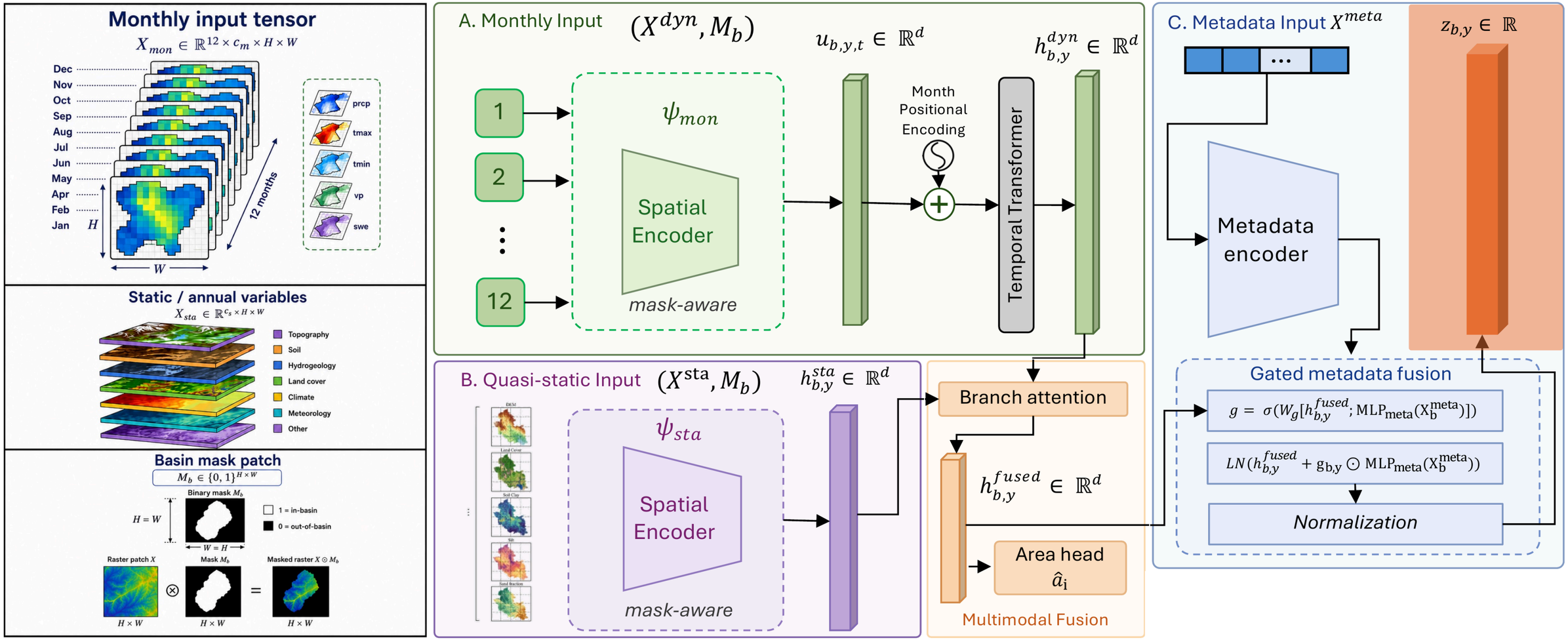
hydrologic relations shape the embedding space

**No observed streamflow** is used

## Output

$z_{b,y} \in \mathbb{R}^d$

# **Study 3:** Scale-aware encoder




**Mask-aware**

Out-of-basin cells never enter the pooling.

**Area head**

Predicts drainage area, so physical basin size is kept.

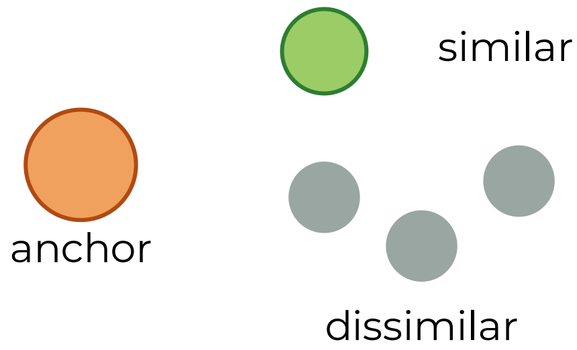
**Gated metadata**

Basin geometry is added through a learned gate.

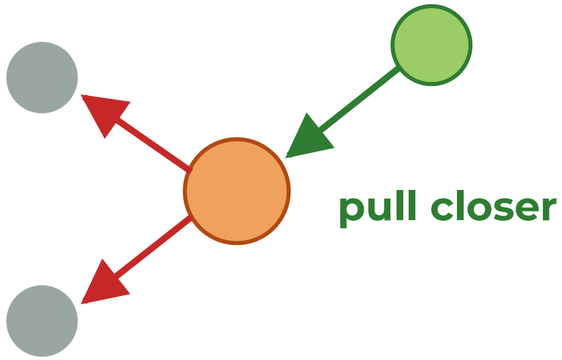
# **Study 3:** Contrastive learning

Learn an embedding space by pulling related samples together and pushing unrelated samples apart.

## Original and candidate samples

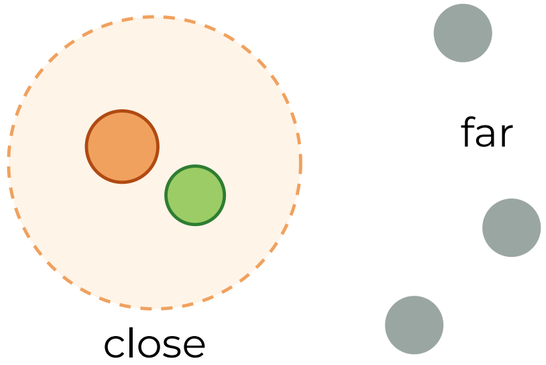



## Contrastive learning rule




maximize similarity to positives,
minimize similarity to negatives

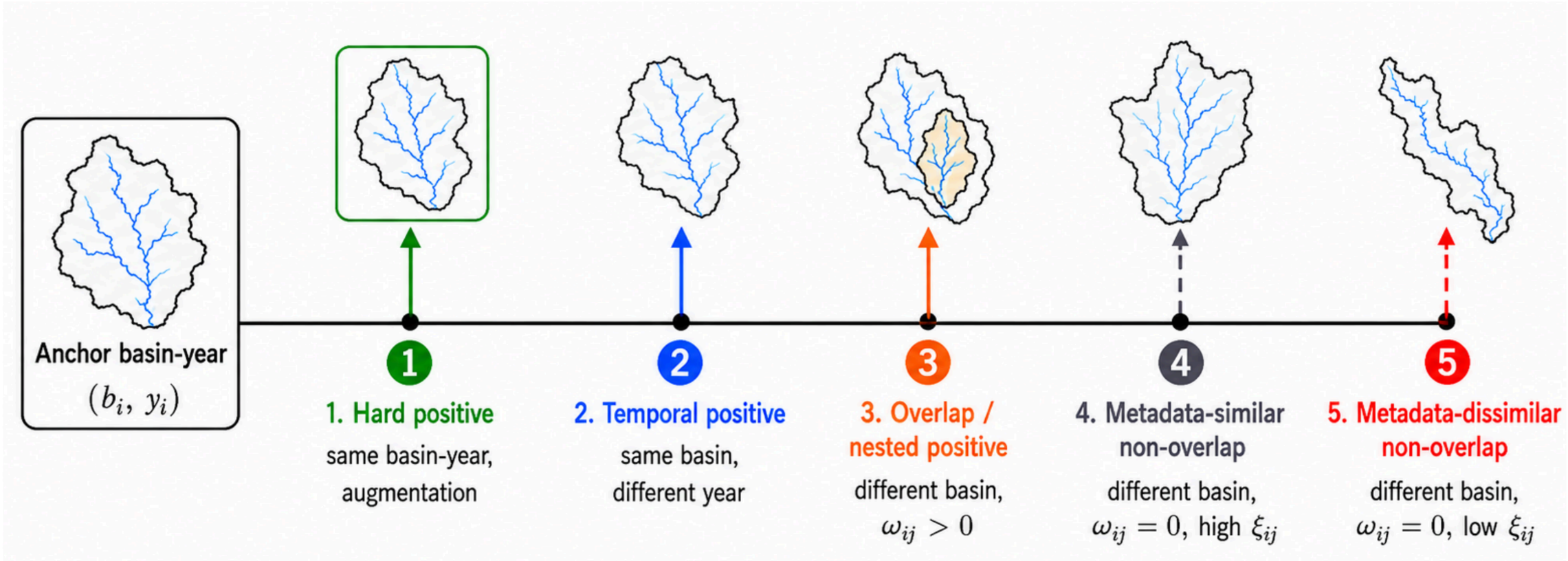
## Learned embedding space




The standard rule treats only two views of the same sample as related. We extend it to **hydrologically meaningful relations** between basins.

# **Study 3:** Hydrology-aware relations

How do we define hydrologically meaningful relations among basin-years?




**Temporal positive**

weight decays with the gap in years, $e^{-|y_i - y_j|/\tau_y}$

**Overlap / nested positive**

weighted by overlap, $\omega_{ij} = \frac{|G_i \cap G_j|}{\min(|G_i|, |G_j|)}$

**Negatives**

pushed apart more weakly when basin metadata are similar

We use **time**, **space**, and **basin similarity** to structure the embedding space.

# **Study 3:** Hydrology-aware contrastive loss

$$\mathcal{L} = \mathcal{L}_{con} + \lambda_n \mathcal{L}_{neg} + \lambda_a \mathcal{L}_{area}$$

## Relation attraction

$$\mathcal{L}_{con}$$

- **Pull** hard, temporal, and overlap / nested positives toward the anchor
- Strength set by the relation weights

+

## Negative separation

$$\mathcal{L}_{neg}$$

- **Push** non-overlapping basins below a similarity margin
- Margin is looser for metadata-similar basins

+

## Scale preservation

$$\mathcal{L}_{area}$$

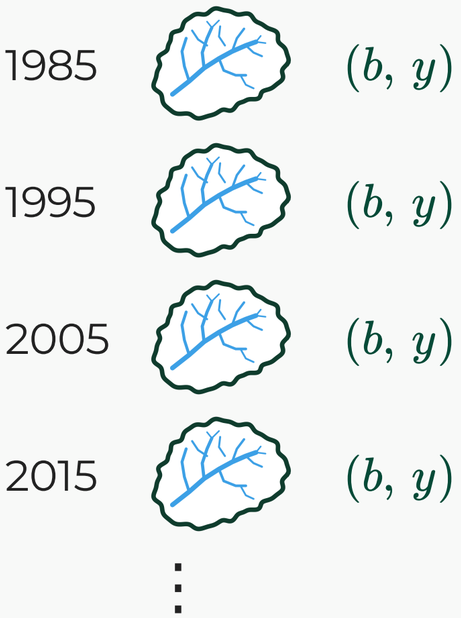
- **Keep** basin size: predict standardized log drainage area
- Smooth-L1, robust to very large basins

Pull related basin-years together, push unrelated basins apart, and keep basin scale.

# **Study 3:** Final product

From basin-year embeddings to one reusable basin descriptor

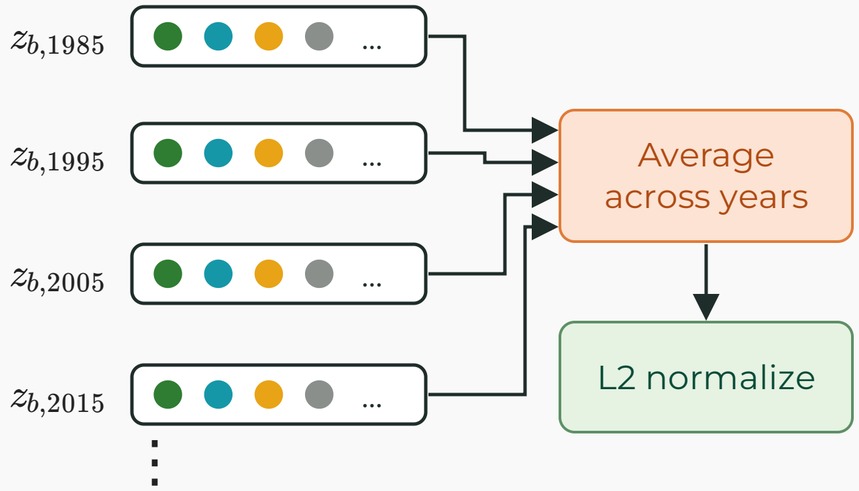
## A basin across years




same basin, different years

## Encode each basin-year

pretrained Basin2Vec encoder

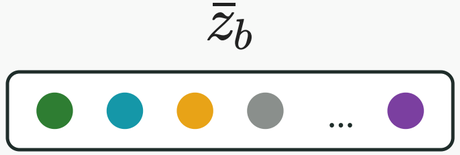



$z_{b,y} \in \mathbb{R}^{64}$

one 64-D embedding per basin-year

$$\bar{z}_b = \text{Normalize}\left(\frac{1}{|Y_b|}\sum_{y \in Y_b} z_{b,y}\right)$$

## Final Descriptor




fixed basin descriptor

one reusable 64-D descriptor per basin

One fixed descriptor is used for the downstream streamflow prediction task.

# **Study 3:** Downstream evaluation setup

Same prediction setup; only the basin descriptor changes.

$$\hat{Q}_{b,t} = F_\eta\big([\, x^{met}_{b,t-L+1}, \ldots, x^{met}_{b,t} \,],\ c_b\big), \qquad L \in \{120,\ 365\}$$

## Daily weather $x^{met}$

Daymet precipitation, shortwave radiation, max and min temperature, vapor pressure

no observed streamflow as input

## Basin descriptor $c_b$ · frozen

**Basin2Vec**

- CAMELS attributes (27)
- AlphaEarth
- TESSERA
- SatCLIP

## Predictor $F_\eta$

- LSTM
- TFT
- TimeXer
- Timer-XL
- iTransformer

Any difference in skill comes from the **representation**, not the model.

# **Study 3:** Generalization experiments

Does the same descriptor generalize across time and across space?

## Temporal generalization

**671** CAMELS-US basins, same basins in both periods

## Zero-shot spatial transfer

**6-fold** ungauged-basin cross-validation

- Held-out basins excluded from training and early stopping
- Train 1990–2004, validate 2005–2014 on the rest
- Held-out basins evaluated over 1990–2014

**Evaluation:** median NSE ↑ and KGE ↑ across basins · 3 seeds

# **Study 3:** Temporal generalization results

Median NSE · future period · L = 365

| Predictor | Basin2Vec | CAMELS attr. | AlphaEarth | TESSERA | SatCLIP |
|---|---|---|---|---|---|
| **LSTM** | **0.782** | 0.747 | 0.641 | 0.662 | 0.719 |
| **TFT** | **0.753** | 0.717 | 0.741 | 0.570 | 0.638 |
| **TimeXer** | **0.651** | 0.502 | 0.630 | 0.573 | 0.603 |
| **Timer-XL** | **0.730** | 0.591 | 0.674 | 0.627 | 0.402 |
| **iTransformer** | **0.627** | 0.555 | 0.559 | 0.576 | 0.464 |

Mean of three seeds. 671 basins, test 2010–2014. Basin2Vec is best in every row.

Average relative gain

**up to 8.3%**

over the best alternative

Consistency

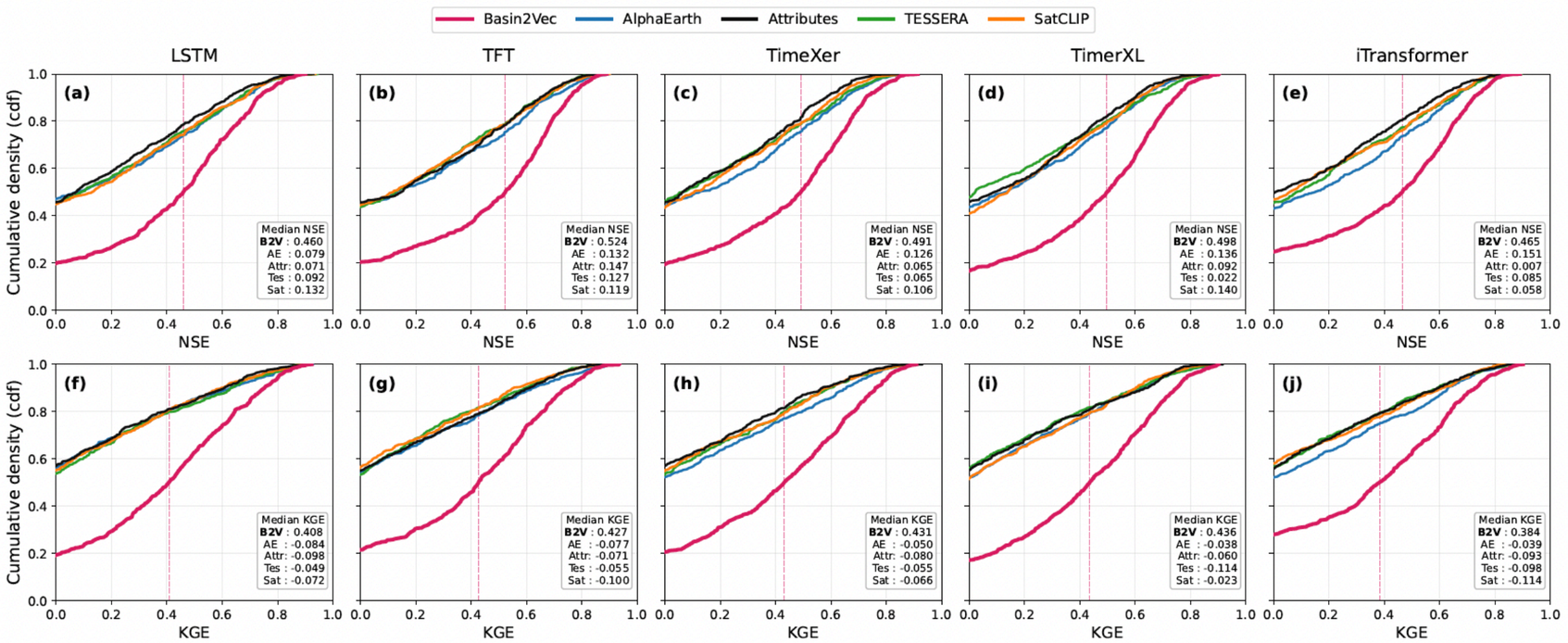
**5 / 5**

predictors: first in NSE and KGE, at L = 120 and 365

Basin2Vec consistently improves future-period prediction across predictors and window lengths.

# **Study 3:** Zero-shot spatial transfer results

## NSE and KGE CDFs · held-out basins




## Median NSE · held-out basins · L = 365

| Median NSE | LSTM | TFT | TimeXer | Timer-XL | iTransformer |
|---|---|---|---|---|---|
| **Basin2Vec** | **0.460** | **0.524** | **0.491** | **0.498** | **0.465** |
| **CAMELS attr.** | 0.071 | 0.147 | 0.065 | 0.092 | 0.007 |
| **AlphaEarth** | 0.079 | 0.132 | 0.126 | 0.136 | 0.151 |
| **TESSERA** | 0.092 | 0.127 | 0.065 | 0.022 | 0.085 |
| **SatCLIP** | 0.132 | 0.119 | 0.106 | 0.140 | 0.058 |

Basin2Vec reaches **3.5×** the median skill of the strongest baseline on unseen basins.

# **Study 3:** Ablation study

Median NSE · LSTM · L = 365

| Variant | Temporal | Spatial |
|---|---|---|
| **Full Basin2Vec** | **0.782** | **0.460** |
| **Scale-aware encoder** | | |
| **w/o monthly branch** | 0.649 | 0.423 |
| **w/o quasi-static branch** | 0.635 | 0.404 |
| **w/o metadata conditioning** | 0.634 | **0.064** |
| **w/o area regularization** | **0.601** | 0.412 |
| **Relation-aware objective** | | |
| **hard positives only** | 0.670 | **0.372** |
| **w/o temporal positives** | 0.621 | 0.418 |
| **w/o overlap / nesting positives** | 0.633 | 0.408 |
| **w/o negative-margin loss** | 0.643 | 0.409 |

## Encoder

Removing **metadata conditioning** collapses unseen-basin skill: **0.460** → **0.064**. The **area term** matters most over time.

## Objective

**Standard** contrastive learning (hard positives only) transfers worst: **0.372** on unseen basins.

Every component helps; the full model is best on every metric.

# **Study 3:** Conclusion

**Study 3 shows that learned basin embeddings outperform the other representation options, especially in unseen basins.**

Established

- **Learned embeddings beat** CAMELS attributes and Earth-observation embeddings.
- First for **all 5 predictors**: up to **8.3%** gain over time, **3.5×** median NSE in unseen basins.
- **Every component helps**, most of all metadata for unseen-basin transfer.

Next Challenge

- One descriptor per basin: one outlet, nothing about the reaches inside.
- Ungauged reaches need the river network explicitly.

**Next: the basin as a directed river network**

**Explore more: Basin2Vec**

Scan, or visit muhammad-talha-mt.github.io/basin2vec

STUDY 4 · ONGOING

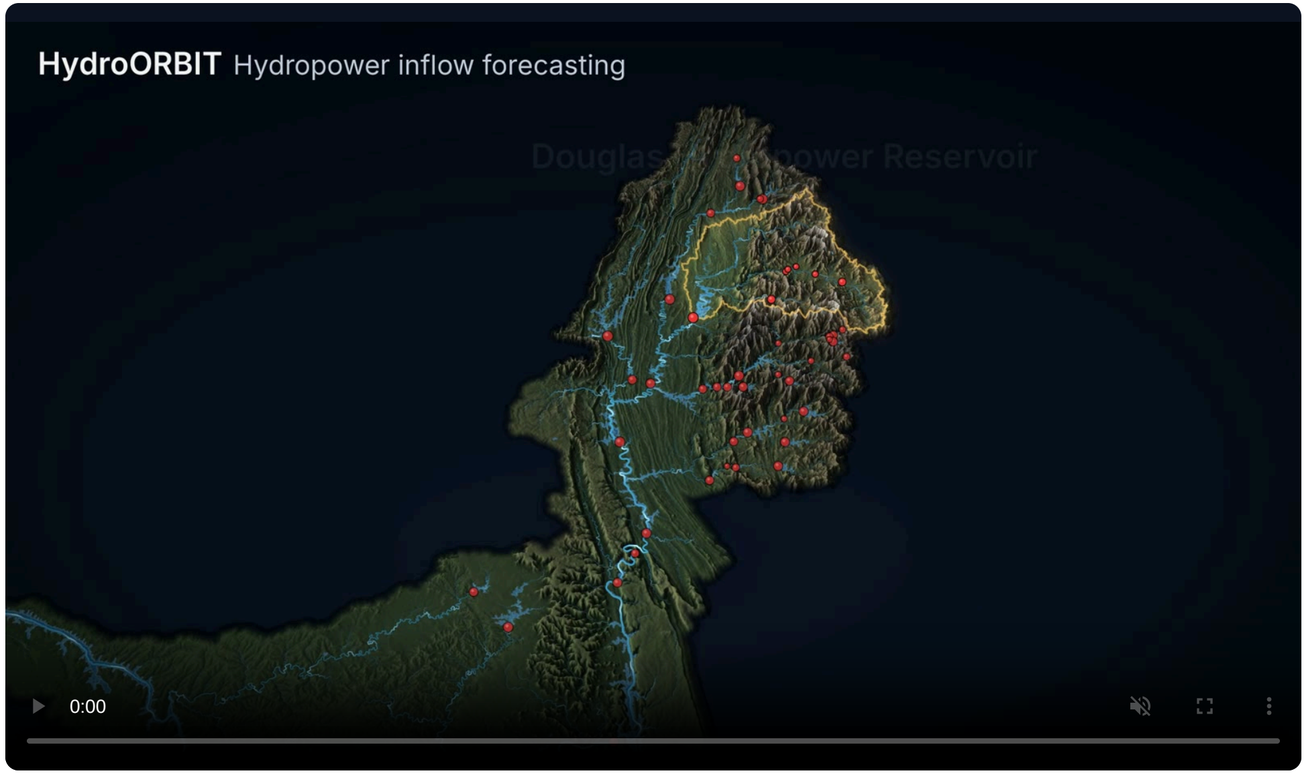
# A River-Network GNN for the Tennessee River Basin

The basin as a directed river network

# **Study 4:** Motivation

**MOTIVATION** Water managers need forecasts where there is no gauge.

## Example: forecasting inflow to a hydropower reservoir




### 1 · The need

Managers need forecasts **days ahead**, also on rivers **with no gauge**.

### 2 · Prior work

GNNs on the Danube: **river topology added almost nothing**.

Kirschstein & Sun, 2024 · every node gauged, 6 h ahead

### 3 · Our setting

Most reaches are **ungauged** and leads reach **10 days**: the network must **carry skill** where no data exist.

**Question:** can a river network carry forecast skill to reaches it has never observed?

# **Study 4:** Problem, forecasting where there is no gauge

**QUESTION THIS ANSWERS** Can one model forecast flow at every reach, and keep its skill at gauges it never saw?

**58,834**

NHDPlus reaches in the Tennessee River Basin (~106,000 km²)

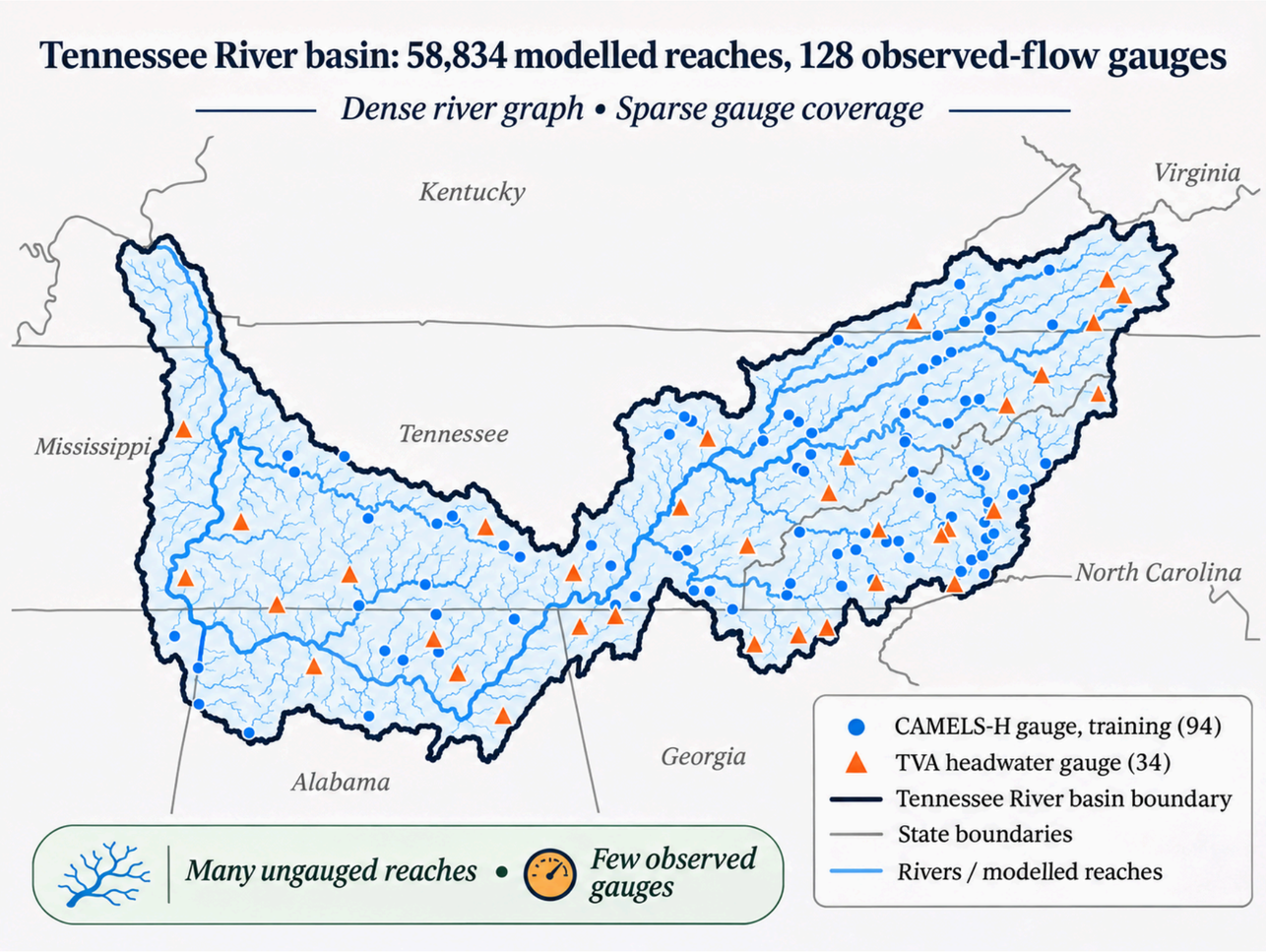
**128**

reaches with a public flow gauge

**> 99.8%**

of reaches are ungauged

## The Tennessee River Basin

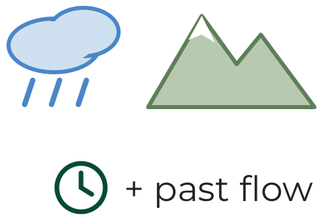



## Forecast task

- **10-day** forecasts, **6-hourly** (40 steps)
- Every reach, one forward pass
- Future input: **issue-time GEFSv12 rain only**

# **Study 4:** Model: reach, neighbours, routing

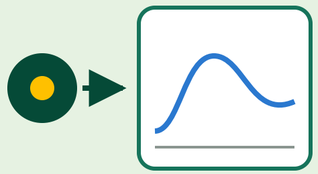
### Inputs



+ past flow

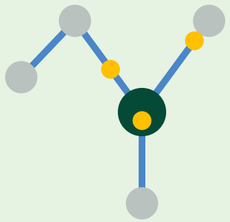
weather · rain forecast
catchment

### 1 · Remember



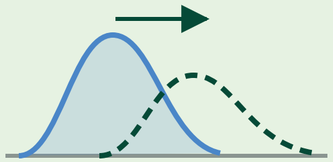
each reach reads
its own history

### 2 · Share



neighbours
exchange information

### 3 · Route



water travels
downstream

### Flow at every reach

**58,834**

reaches

6-hourly, 10 days ahead

**Where a gauge is visible:**
recent observed flow sharpens the forecast, with upper quantiles for floods.

Two-stage training

**A** **Learn from simulation**

$$\mathcal{L}_A = \mathcal{L}\left(\hat{Q},\ Q^{\mathrm{NWM}}\right) \text{ at every reach}$$

**B** **Correct with observations**

$$\mathcal{L}_B = \underbrace{\mathcal{L}_{\mathrm{NSE}}}_{\text{gauges}} + \underbrace{\mathcal{L}_{\mathrm{pinball}}}_{q_{50},\, q_{90},\, q_{99}} + 0.3 \quad \underbrace{\mathcal{L}_{\mathrm{prior}}}_{\text{stay near NWM}}$$

# **Study 4:** Forecast protocol and test design

## One forecast rule for every model

**Issue time · 00 UTC daily**

| **Observed: AORC weather + USGS flow (36 days)** | **Only the GEFSv12 control-member rain forecast** |
|---|---|
| history window | 40 six-hour steps → 10 days |

## Periods

| | |
|---|---|
| **Train** | GEFSv12 reforecast, 2000–2018 |
| **Model selection** | 2019 |
| **Test** | **23 Sep 2020 – 22 Dec 2025: 1,917 daily issues** |

## Two tests at observed gauges

| | |
|---|---|
| **Gauged** | Recent flow shown |
| **Held-out** | 5 folds, recent flow never seen in training |

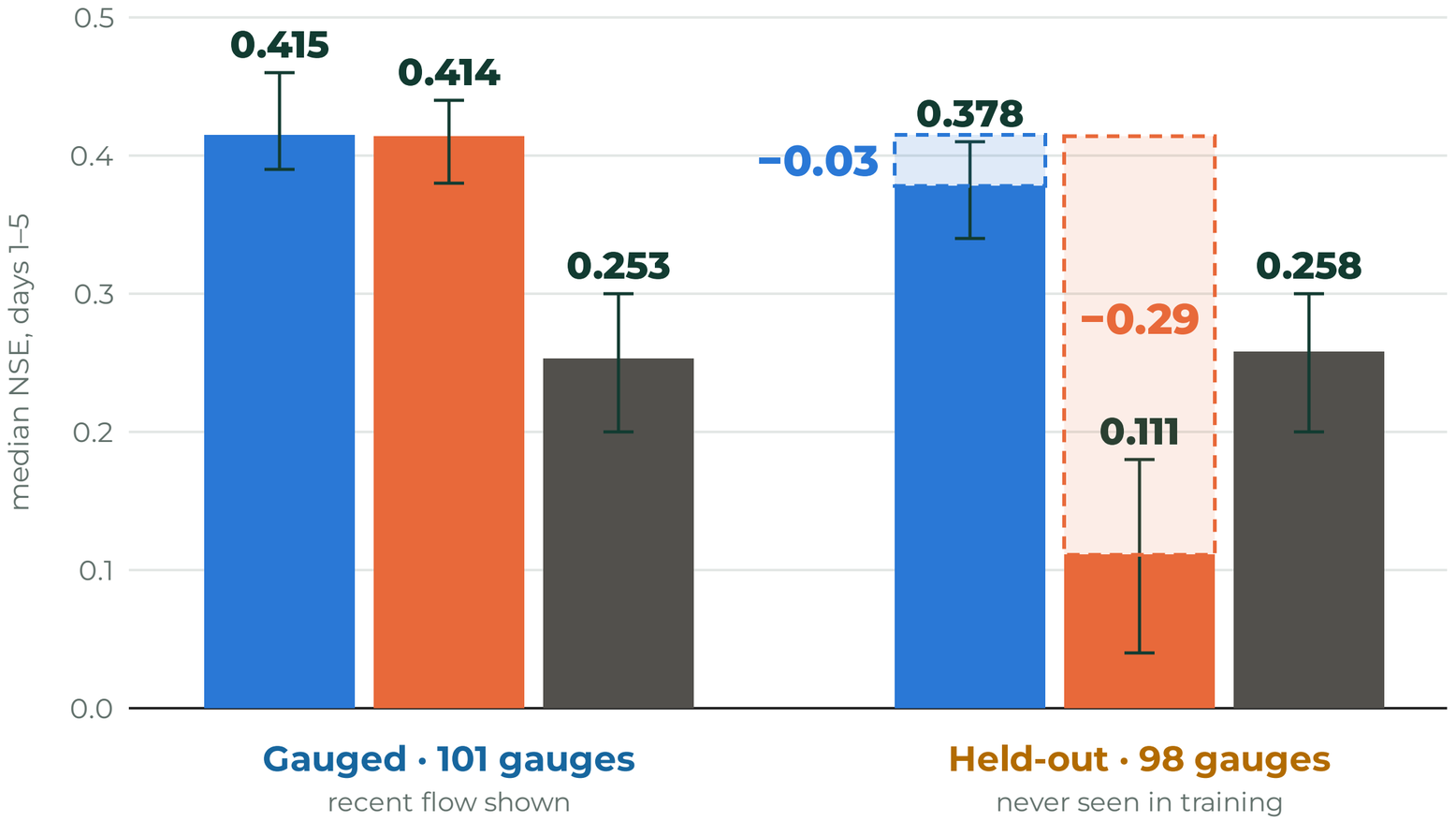
Same issues, gauges, masks, and GEFS rain for GNN and LSTM.

## Baselines

- **Tuned basin LSTM** (256 units), same rain and masks
- **NWM operational forecast**: a reference; own forcing and assimilation

Score: per-gauge mean NSE over 24, 48, 72, 96, 120 h ("days 1–5"), median over gauges, 95% gauge-bootstrap interval.

# **Study 4:** Key result, skill at gauges the model never saw




**GNN** **LSTM** (tuned)

**NWM** operational

**Gauged**

GNN and LSTM **tie**: 0.415 vs 0.414

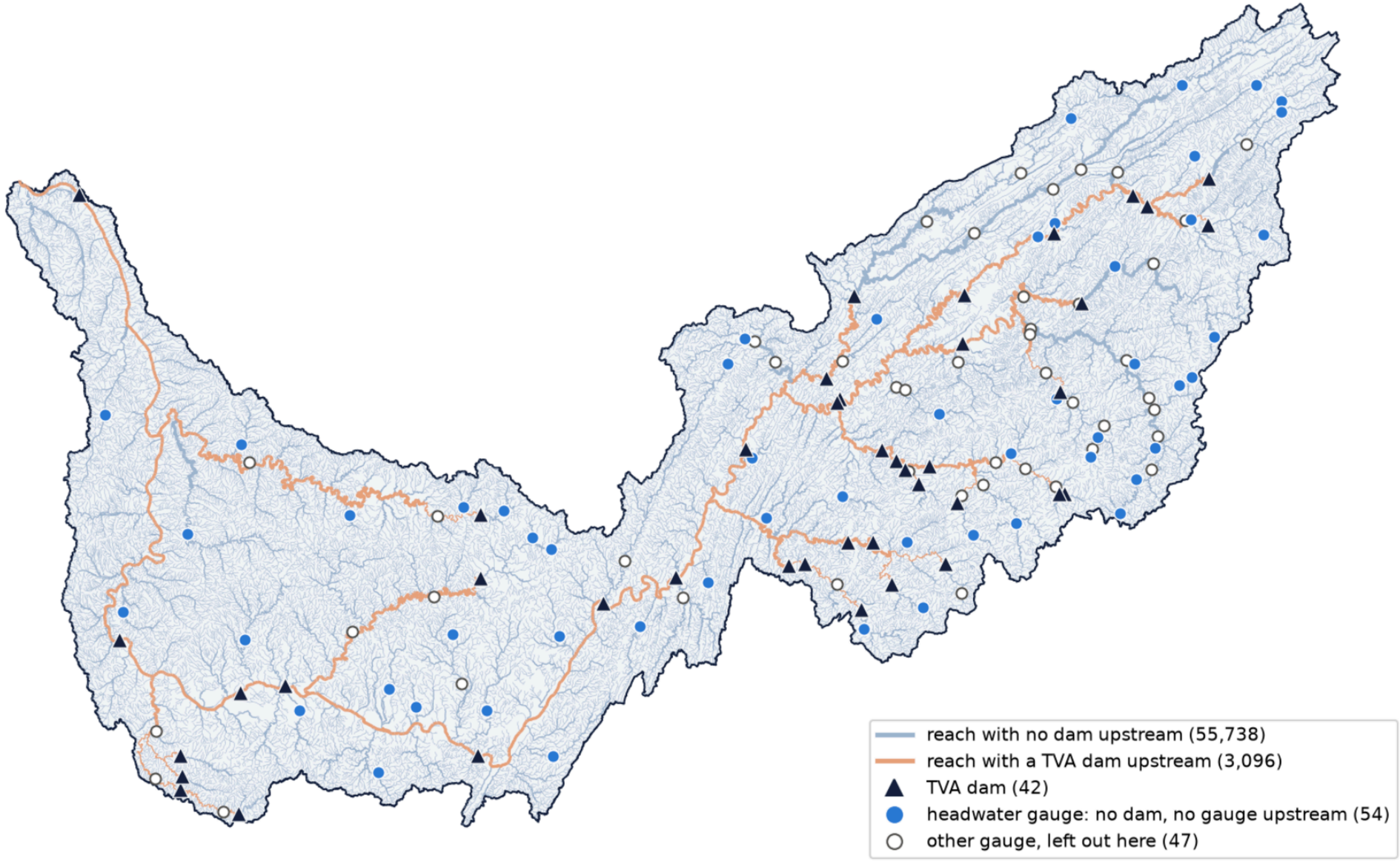
**Held-out**

GNN **0.378** (0.34–0.41)
LSTM 0.111 (0.04–0.18)
NWM 0.258

Whiskers: 95% gauge-bootstrap intervals. NWM assimilates USGS observations and may use held-out gauges, so it is a reference, not a controlled comparison.

**Losing the gauge costs the GNN 0.03 NSE, the LSTM 0.29.**

# **Study 4:** A regulated basin: the headwater gauges




**42**

TVA dams regulate the main stems

**54**

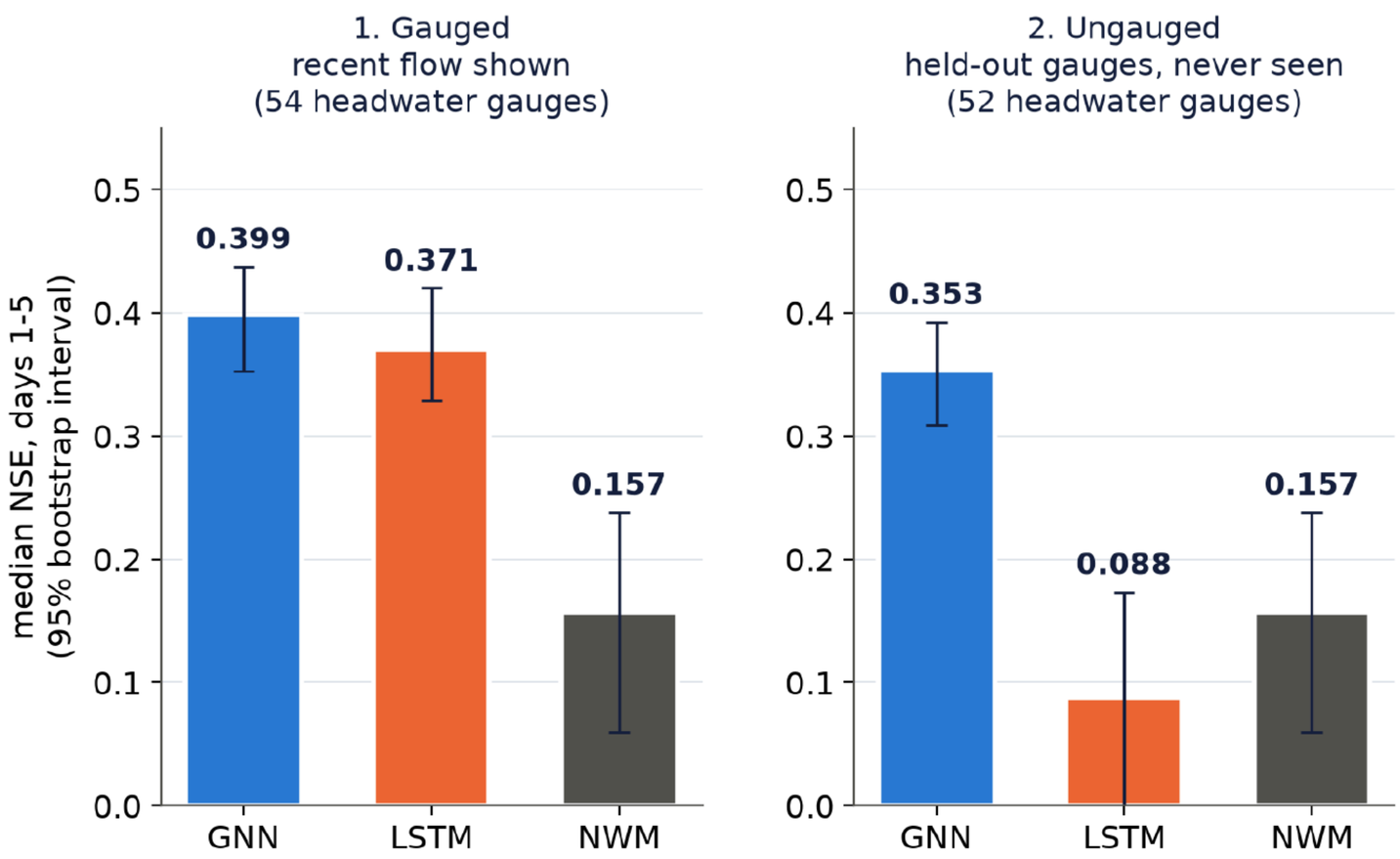
headwater gauges: no dam, no gauge upstream

**95%**

of reaches have no dam upstream

# **Study 4:** Headwater basins without dams

Median NSE, days 1–5




Gauged

**94%**

of headwater gauges where the GNN beats NWM

Held-out

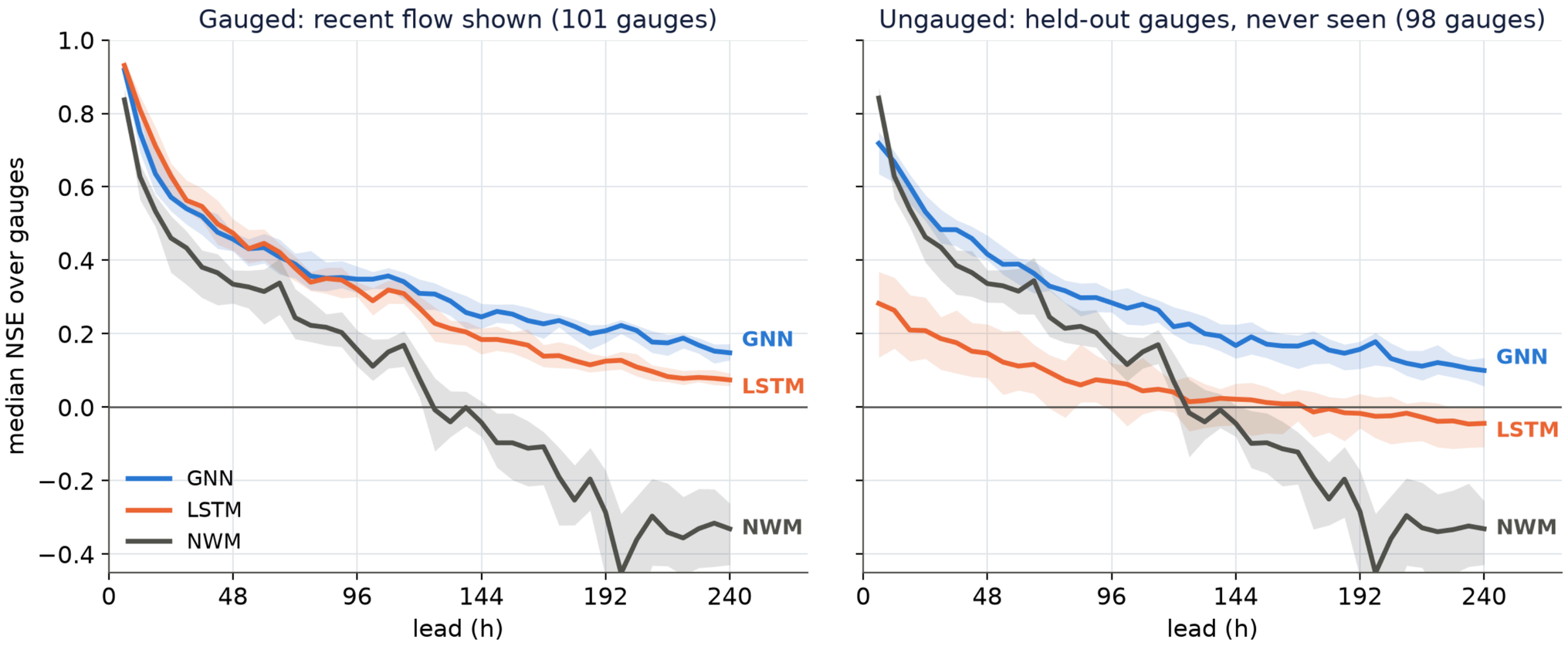
**90%**

of unseen gauges where the GNN beats the LSTM (87% vs NWM)

Without dams, the GNN is **ahead at gauges** and far ahead at unseen gauges.

# **Study 4:** Lead time

## Median NSE by lead time, 6 h → 10 days



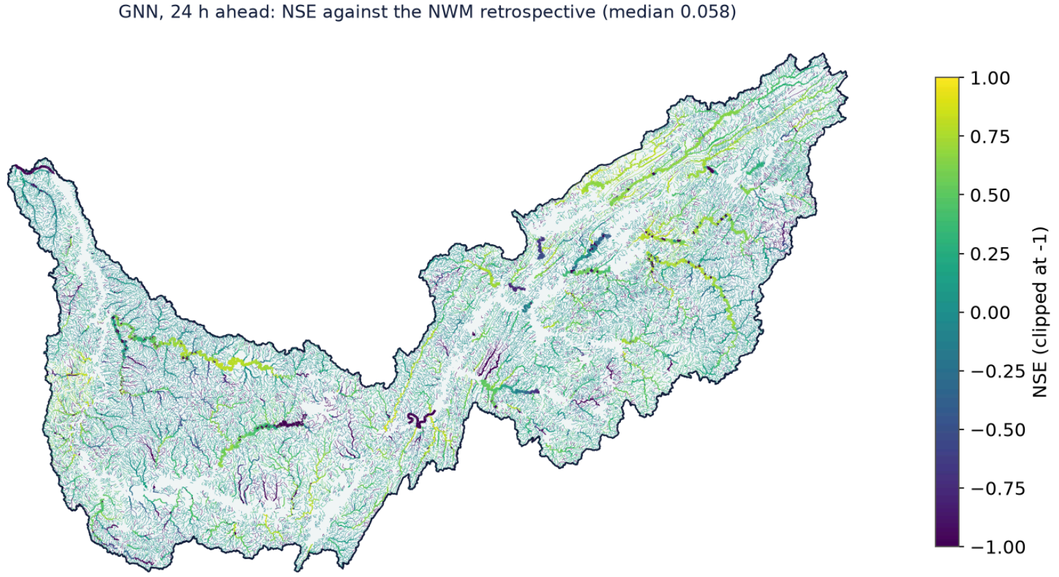

### Gauged

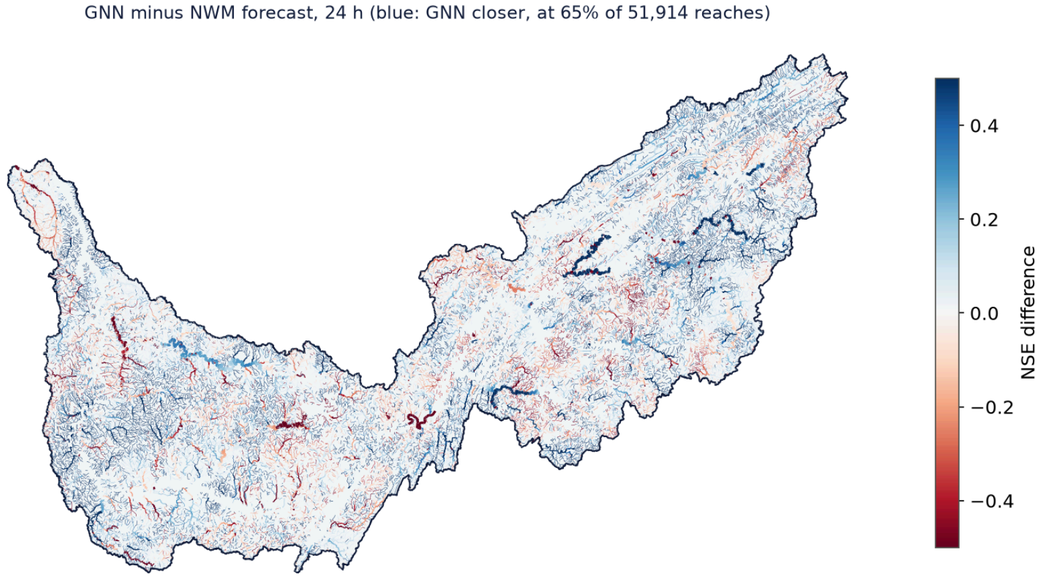
LSTM leads the first day. The **GNN leads from ~3 days**.

### Held-out

The **GNN beats the LSTM at every lead**. NWM wins only at 6 h.

# **Study 4:** Every reach, 24 h ahead




## How often is the GNN closer?

**vs NWM · 24 h**

**65%**

of reaches: the GNN is closer

**vs NWM · 72 h**

**70%**

of reaches: the GNN is closer

**vs LSTM · 24 h**

**88%**

of reaches: the GNN is closer

The reference is a simulation, not observations.

# **Study 4:** Conclusion

**A river-network representation transfers to withheld gauges** under a realistic protocol.

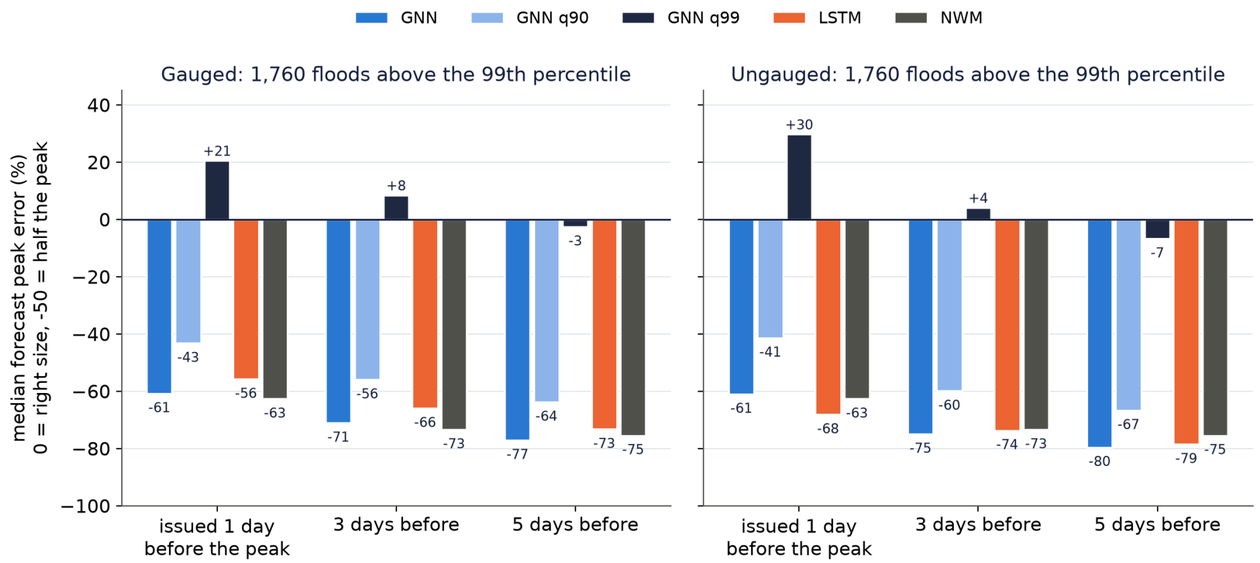
## Where the GNN helps

- **Ungauged reaches:** keeps skill where no gauge was seen.
- **Longer leads:** strongest beyond ~3 days.
- **Natural-flow headwaters:** ahead even at gauges.

## Limitations

- Floods: peaks too low; extremes missed.
- Scale: one basin; other regions untested.
- Attribution: graph, routing, or prior?

## Limitation: flood peaks are too low




Peaks above the 99th percentile: **56–80% too low** for every model. Hurricane Helene (28–45×) was missed by all.

**Next: tokenize the basin and pretrain representations at scale**

# What's next at ORNL

PROJECT

## HydroORBIT

From one basin to a **pretrained, national** forecasting model

with Dan Lu, Xiao Wang, Vinh Tran

1

### Tokenize the basin

Fields, river network and gauges become one shared input.

2

### Pretrain at scale

Learn once from simulation and sparse observations, on Frontier.

3

### Deliver forecasts that transfer

Inflow and flood forecasts at ungauged reaches, with uncertainty.

**First, close the open questions:** why the network transfers, flood peaks, transfer beyond Tennessee.

# Summary

Hydrologic generalization depends on **how we represent the basin**.

## Established

- **Spatial fields** beat lumped inputs, most in unseen basins.
- **BASINGRID** standardizes distributed data at national scale.
- **Learned basin embeddings** are reusable across predictors and transfer to unseen basins.
- **A river network** transfers to withheld gauges.

## To be established

- **Flood peaks** along the river network.
- **Transfer beyond one region**, at national scale.
- **Pretrained, reusable** representations (HydroORBIT).

**Next: tokenize the basin and pretrain representations at scale.**

# Thank you

## Questions?

**Access these slides**
muhammad-talha-mt.github.io/ORNL_Seminar

**My profile and papers**
muhammad-talha-mt.github.io

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